SIX-FUNCTOR FORMALISM FOR SOLID SYNTOMIC COHOMOLOGY OF DERIVED P-ADIC FORMAL SCHEMES

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Allowed standard prerequisites (editorial).

- Prismatic cohomology, analytic stacks, solid modules and abstract six-functor formalisms. Original prerequisite statements, author constructions and complete Paperbib.bib retained.

Introduction

Grothendieck suggested that a cohomology theory should come together with its counterparts, homology, compactly supported cohomology, and Borel-Moore homology. Furthermore, these invariants are supposed to interact with one another in terms of certain maps and module structures, and all of this structure should have a fibered interpretation. Nowadays this structure has a precise interpretation and runs under the name *six-functor formalism* (see for example [HM24], [Sch25] and [CD09]). In this text we construct this structure for syntomic cohomology of p-adic formal schemes as defined in [BS19] [BMS18] and [BL22a].

Theorem (**A** see 4.1.3)**.** *There is a six-functor formalism*

$$X \mapsto F\text{-}Gauge^{\square}_{\blacktriangle}(X)$$

on the category of (derived) p-adic formal schemes satisfying the following:

(A) Morphisms locally of finite type are !*-able.*
(B) Étale morphisms are cohomologically étale.
(C) Any finite composition of closed immersions and proper morphisms[1] *is weakly cohomologically proper.*
(D) The functor $(_)^{syn,\square}$ *preserves étale covers and sends Zariski covers to open covers of analytic stacks. In particular the functors* $F\text{-}Gauge^{\square}_{\blacktriangle}(_)^*$ *and* $F\text{-}Gauge^{\square}_{\blacktriangle}(_)^!$ *are étale sheafs.*
(E) It admits Tate twists. That is for any p-adic formal scheme X *the object*

$$\mathcal{O}_{X^{syn}}(-1) := cof(\mathcal{O}_{X^{syn}} \to f_*\mathcal{O}_{(\mathbf{P}^1)^{syn}})$$

is $\otimes$*-invertible and its inverse identifies with*

$$\mathcal{O}_{X^{syn}}(1) \simeq \mathcal{O}_{X^{syn}}\{1\}[-2]$$

a shift of the Breuil-Kisin twist.

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[1] In this note proper morphisms are locally of finite presentation.

(F) *Any smooth morphism is cohomologically smooth. Furthermore, for such a smooth morphism $f\colon X \to S$, we have an identification*

$$f^! \mathcal{O}_{S^{syn}} := \omega_f \simeq \mathcal{O}_{X^{syn}}(d) \simeq \mathcal{O}_{X^{syn}}\{d\}[-2d]$$

of the dualizing sheaf, where d is the relative dimension of f.

(G) *The dualizable*[2] *objects identify*

$$(F\text{-}Gauge^{\square}_{\mathbb{\Delta}}(_))^{dual} \simeq Perf((_)^{syn})$$

with perfect F-Gauges as defined in [BL22b][6.1]. In particular, there is a functorial identification

$$R\Gamma^{BMS}_{syn}(_, \mathbf{Z}_p(n)) \simeq Hom_{(_)^{syn}}(\mathcal{O}_{(_)^{syn}}, \mathcal{O}_{(_)^{syn}}\{n\})$$

of the mapping spectrum with the syntomic cohomology of p-adic formal schemes as defined in [BL22a].

One reason such a theorem is interesting is that, for any smooth morphism $f\colon X \to S$ of relative dimension d, it produces a relative compactly supported cohomology

$$f_! \mathcal{O}_{X^{\mathrm{syn}}} \in \text{F-Gauge}^{\square}_{\mathbb{\Delta}}(S)$$

which gives a Poincaré duality pairing. That is, we obtain the following corollary.

Corollary. *The identification in (F) induces a trace map*

$$tr\colon f_! \mathcal{O}_{X^{syn}}\{d\}[-2d] \to \mathcal{O}_{S^{syn}}$$

such that the composition

$$f_* \mathcal{O}_{X^{syn}} \otimes_S f_! \mathcal{O}_{X^{syn}}\{d\}[-2d] \xrightarrow{\cup} f_! \mathcal{O}_{X^{syn}}\{d\}[-2d] \xrightarrow{tr} \mathcal{O}_{S^{syn}}$$

is adjoint to an isomorphism

$$f_* \mathcal{O}_{X^{syn}} \simeq (f_! \mathcal{O}_{X^{syn}}\{d\}[-2d])^{\vee}$$

in F-Gauge$^{\square}_{\mathbb{\Delta}}(S)$.

Another reason this theorem is interesting lies in the nature of Syntomic cohomology, which we now recall very briefly.

One way to understand syntomic cohomology is, following Bhatt-Lurie [BL22a] [BL22c] and Drinfeld [Dri24], via the stacky approach. That is, to a p-adic formal scheme X, one associates a stack X^{syn}, such that quasi-coherent sheaves on this stack recover syntomic cohomology.

In order to obtain our theorem, we interpret these stacks in the world of analytic stacks introduced by Clausen-Scholze [CS24]. That is to a p-adic formal scheme X we now associate an analytic stack

$$X^{\mathrm{syn},\square} \in \mathrm{AnStack},$$

which we call the *solid Syntomification* of X, such that its category of quasi-coherent sheaves recovers syntomic cohomology. This has (among others) two advantages:

One is that on quasi-coherent sheaves of analytic stacks it is easier[3] to define a well-behaved compactly supported cohomology. This is crucial to obtain the above theorem.

[2] There is also a version containing the full category of prismatic F-Gauges 4.3.6

[3] Maybe possible would be the better wording here.

Another advantage is that the category of analytic stacks is quite huge and contains a priori very different looking stacks. In particular, in this world, it is easier to find relations between different looking geometries.

This latter advantage could be particularly useful for the syntomification, as it philosophically is supposed to parametrize well-behaved p-adic cohomology theories[4] and **Theorem A** gives a way to apply this idea. Namely, given a map

$$\mathcal{B} \to \mathbf{Z}_p^{\mathrm{syn},\square}$$

of analytic stacks. Then it is not hard to see that assigning to a p-adic formal scheme X the category

$$\mathcal{D}(\mathcal{B} \times_{\mathbf{Z}_p^{\mathrm{syn},\square}} X^{\mathrm{syn},\square})$$

can be extended to a six-functor formalism that satisfies (A)-(F). We record one example of this.

Theorem (**B** see 4.3.10)**.** *There is a six-functor formalism*

$$X \mapsto \mathcal{D}(X_\eta^{\acute{e}t,\square})$$

on p-adic formal schemes over $\mathbf{Z}_p^{cyc}$, which satisfies analogous assertions to (A)-(F). Furthermore, the dualizable objects naturally identify

$$\mathcal{D}((_)_\eta^{\acute{e}t,\square})^{dual} \simeq \mathcal{D}^b_{lisse}((_)_\eta, \mathbf{Z}_p)$$

with the category of lisse étale sheaves on the generic fiber.

The way we prove the Poincaré duality is somewhat disconnected from these geometric ideas, in the sense that we give a general strategy to prove this duality in six-functor formalisms. This should be applicable in many other situations and is highly motivated by the work of Zavyalov [Zav23], Tang [Tan26] [Tan] and Annala-Hoyois-Iwasa [AHI25] [AHI24]. That is, we explain the following theorem.

Theorem (**C** see 1.6.5)**.** *Given a six-functor formalism $\mathcal{D}$ on the category of schemes, which satisfies the following (and admits the following structures):*

- *All étale morphisms of schemes are cohomologically étale.*
- *All proper morphisms are (weakly) cohomologically proper.*
- *$\mathcal{D}^*$ is a Nisnevich sheaf.*
- *It admits an additive orientation.*
- *It satisfies elementary blow-up excision.*

Then any smooth morphism $f\colon X \to S$ of relative dimension d will be cohomologically smooth and we have a canonical identification

$$f^!(\mathbf{1}_S) \simeq \mathbf{1}_X(d)$$

of the dualizing sheaf with the Tate twist[5].

Here an additive orientation is given by a theory of first Chern classes together with a projective bundle formula isomorphism. This in practice reduces to a simple computation of cohomology of projective spaces. Also elementary blow-up excision

[4] See for example [LM25] for some precise statement.

[5] Here the Tate twist is given by the homology of pointed $\mathbf{P}^1$. In the text we will switch to $(d)[-2d]$.

reduces to a computation of cohomology of blow-ups of zero-sections in affine space and in practice should often even reduce to the projective bundle formula.

In the text **Theorem C** is not only formulated for six-functor formalisms on the category of schemes. Instead, we tried to axiomatize what properties smooth morphisms need to satisfy to make the theorem work. In particular, this should logically be applicable to geometric settings like complex analytic spaces, rigid spaces or Berkovich spaces as well.

Organization of the Text. In 1 we prove **Theorem C** and give the necessary axiomatizations and constructions.

In 2 we explain how to understand formal stacks in terms of analytic stacks. Most importantly, we give a criterion for properness of analytic stacks coming from formal stacks 2.3.3.

In 3 we recall some aspects of the stacky approach to syntomic cohomology and define the solid Syntomification.

In 4 we collect the proof of **Theorem A**.

Notations and conventions. We will use the following notations and conventions.

- We will use the term *category* to mean what, often in the literature, is called an ∞-category.
- A ring will be an animated commutative ring.
- Many topoi in this text are "big topoi". That is, they are formally not topoi because of size issues. We will systematically ignore this issue, as we will never use that they are topoi but just that they practically behave like those in the following sense. To deal with this issue, one defines a κ-small version of this topos for each uncountable strong limit cardinal κ and then takes the colimits over those κ in categories. In particular, most arguments will happen in the κ-small subcategory for some κ and thus in some topos.
- We will use the homological conventions to speak about (co)homology groups, t-structures and suspensions.

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1. A remark on smoothness in six-functor formalisms

1.1. **Recollections on six-functor formalisms.** We recall some facts and definitions in the context of six-functor formalisms. More detailed discussions can be found in [HM24] [GR17] [CD09] [Sch25].

1.1.1. Consider a pair $(\mathcal{C}, \mathcal{C}_E)$ where $\mathcal{C}$ is a category and $\mathcal{C}_E \subset \mathcal{C}$ a wide subcategory satisfying the following:

(a) Morphisms in $\mathcal{C}_E$ are closed under pullbacks along morphisms in $\mathcal{C}$.
(b) $\mathcal{C}_E$ admits pullbacks and the inclusion $\mathcal{C}_E \subset \mathcal{C}$ preserves those.

Such data is called a *geometric setup* and given such data we can construct the category

$$\mathrm{Span}(\mathcal{C}, \mathcal{C}_E)$$

which informally can be described as follows (see [HM24][2.2] for an honest construction):

- Objects are given by the objects in $\mathcal{C}$.
- A morphism from X to Y is given by a span

$$\begin{array}{ccc} & Z & \\ \swarrow & & \searrow \\ X & & Y \end{array}$$

  where the right leg lives in $\mathcal{C}_E$. To compose such spans, one takes fiber products of the inner cospan.

Assuming that $\mathcal{C}$ admits finite products, the category $\mathrm{Span}(\mathcal{C}, \mathcal{C}_E)$ can be equipped with a symmetric monoidal structure induced by the cartesian product in $\mathcal{C}$.

The following definition is taken from [HM24].

Definition 1.1.2. Given a geometric setup $(\mathcal{C}, \mathcal{C}_E)$, such that $\mathcal{C}$ admits finite products. A *six-functor formalism* on $(\mathcal{C}, \mathcal{C}_E)$ is a lax symmetric monoidal functor

$$\mathcal{D} \colon \mathrm{Span}(\mathcal{C}, \mathcal{C}_E) \to \mathrm{Pr}^L.$$

Where Pr^L denotes the category of presentable categories with colimit-preserving functors as morphisms and equipped with the Lurie-tensor product.

Remark 1.1.3. Using [HM24][3.4] we can and will assume that $\mathcal{C}$ is a topos, such that $\mathcal{D}^*$ defines a sheaf on this topos and $\mathcal{D}^!$ descends along effective epimorphisms in $\mathcal{C}_E$.

1.1.4 (**The category of Kernels**)**.** Given a six-functor formalism and an object $X \in \mathcal{C}$, we obtain a restricted six-functor formalism

$$\mathcal{D} \colon \mathrm{Span}(\mathcal{C}(X)) \to \mathrm{Pr}^L$$

where we write $\mathrm{Span}(\mathcal{C}(X)) := \mathrm{Span}((C_E)_{/X}, (C_E)_{/X})$. Furthermore, $\mathcal{D}$ gives a commutative algebra object in $\mathrm{Fun}(\mathrm{Span}(\mathcal{C}(X)), \mathrm{Pr}^L)$ where the later category is equipped with the day convolution symmetric monoidal structure. We then write

$$\mathcal{K}_X := \mathrm{Mod}_{\mathcal{D}}(\mathrm{Fun}(\mathrm{Span}(\mathcal{C}(X)), \mathrm{Pr}^L))$$

and call it the *category of kernels* of X.

This can be interpreted as a 2-category, where any object $X \in \mathcal{C}_E$ give rise to an object $[X] \in \mathcal{K}_*$, 1-morphisms $[X] \to [Y]$ are described by object in $\mathcal{D}(X \times Y)$ and 2-morphism between those are described by morphisms in $\mathcal{D}(X \times Y)$.

Remark 1.1.5. In this text, the category of kernels will only be used implicitly. For a more detailed description and construction, we refer to [HM24][4] [Sch25][V].

We will use the following notions [HM24][4] [Cam24][3.1.].

Definition 1.1.6. Let $f\colon X \to S$ be a morphism in $\mathcal{C}_E$ and $\mathcal{E} \in \mathcal{D}(X)$.

(a) $\mathcal{E}$ is called *f-suave*, if $\mathcal{E}\colon [X] \to [S]$ is left adjoint in $\mathcal{K}_S$.
(b) $\mathcal{E}$ is called *f-prim*, if $\mathcal{E}\colon [S] \to [X]$ is left adjoint in $\mathcal{K}_S$.
(c) f is called *cohomologically smooth*, if $\mathbf{1}_X$ is f-suave and $\mathrm{D}_f(\mathbf{1}_X) = f^!(\mathbf{1}_S)$ is $\otimes$-invertible.
(d) f is called *cohomologically co-smooth*, if $\mathbf{1}_X$ is f-prim and $p_{2,*}\Delta_!(\mathbf{1}_X) = \mathrm{P}_f(\mathbf{1}_X)$ is $\otimes$-invertible.
(e) f is called *cohomologically étale*, if it is n-truncated for some finite n, $\mathbf{1}_X$ is f-suave and the diagonal Δ_f is cohomologically étale or an isomorphism.
(f) f is *weakly cohomologically proper*, if $\mathbf{1}_X$ is Δ_f-prim, there exist some (non canonical) isomorphism $\mathrm{P}_{\Delta_f}(\mathbf{1}_X) \simeq \mathbf{1}_X$ and $\mathbf{1}_X$ is f-prim.

Remark 1.1.7. All of the above notions admit dual versions, which will not appear in this text. In particular, all "étale" morphisms we will consider will be truncated. In contrast, almost no "proper" morphisms we will consider will be truncated.

Remark 1.1.8. All the above notions are stable under base change and composition.

We will use the following notions of coverings. Recall the notion of descendable algebra from [Mat16][3.3.].

Definition 1.1.9. Consider a morphism $f\colon X \to S$ in $\mathcal{C}_E$. Then we will say:

(a) f is a smooth cover, if it is cohomologically smooth and f^* is conservative.
(b) f is a descendable cover, if it is cohomologically co-smooth and
$$f_*\mathbf{1}_X \in \mathcal{D}(S)$$
is descendable.
(c) f is a proper descendable cover, if it is a descendable cover and weakly cohomologically proper.

Remark 1.1.10. For all the above covers, one gets that $\mathcal{D}^*$ as well as $\mathcal{D}^!$ descend along them [HM24][4.7.1, 4.7.4].

To end the section, we record the following stability properties.

Proposition 1.1.11. *We have the following.*

(a) Being cohomologically smooth is $\mathcal{D}^*$-local on the target and smooth local on the source.
(b) Being cohomologically étale is $\mathcal{D}^*$-local on the target and étale local on the source.
(c) Being weakly cohomologically co-smooth is $\mathcal{D}^*$-local on the target and proper descendable local on the source.

(d) *Being a smooth cover is stable under composition, base change, and $\mathcal{D}^*$-local on the target.*
(e) *Being a descendable cover is stable under composition, base change, and descendable local on the target.*

PROOF. Assertion (a) is explained in [HM24][4.4.8+4.5.8(i)] (see also [HM24][4.5.12] and [Lur17][4.4.1.11]). Assertion (b) is [HM24][4.6.3](ii). Assertion (c) is [Cam24][3.1.14] or [HM24][4.4.8] and [HM24][4.7.5](ii). The other assertions are now clear if one has [Mat16][3.24] in mind. □

Remark 1.1.12. Note that, if we, for a co-smooth map $f\colon X \to S$, know that the diagonal is weakly cohomologically proper. Then also f is weakly cohomologically proper.

Remark 1.1.13. If the six-functor formalism is constructed from a suitable decomposition of E [HM24][3.3.2]. That is, the construction comes with two classes (I, P) of étale and proper maps, which are cohomologically étale resp. cohomologically proper by construction. Then if a map $f\colon X \to S$ locally on S for a $\mathcal{D}^*$-cover lies in P, it is weakly cohomologically proper. This is explained in [Cam24][3.1.21].

1.2. **Additively oriented six-functor formalisms.** In what follows, we will incorporate some geometry into our six-functor formalism. Most importantly, we implement the so-called Projective bundle formula. For this, we fix a six-functor formalism $\mathcal{D}$.

Definition 1.2.1. We will say a geometric setup $(\mathcal{C}, \mathcal{C}_E)$ is *geometrized*, if it comes with functor

$$\mathrm{Sm}_B^{\mathrm{sep}} \to \mathcal{C}_E$$

from the category of separated smooth schemes over some (possibly derived) scheme B, which preserves those finite limits, which are also finite limits in the category of all (derived) schemes.

Furthermore, a six-functor formalism $\mathcal{D}$ on a geometrized geometric setup is called *geometric*, if:

- For any object $S \in \mathcal{C}$ the category $\mathcal{D}(S)$ is stable.
- Any étale morphism in Sm_B is cohomologically étale with respect to $\mathcal{D}$.
- Any proper morphism in Sm_B is weakly cohomologically proper with respect to $\mathcal{D}$.
- Any Nisnevich covering gets sent to a $\mathcal{D}^*$-cover.

Remark 1.2.2. We will normally not write the functor $\mathrm{Sm}_B \to \mathcal{C}$. That is, for example, we will write $\mathbf{A}^n$ for the n-dimensional affine space seen as an object in $\mathcal{C}$ via this functor. We can also define the affine space over an arbitrary object $X \in \mathcal{C}$ via base change.

Remark 1.2.3. The Nisnevich topology is famously used in motivic homotopy theory and sits in between the Zariski and the étale topology. The étale topology would be enough for all applications in this text. We just chose this definition for the sake of generality.

For the rest of the section, we assume that the fixed six-functor formalism is geometric[6].

Definition 1.2.4. Let us write $f\colon \mathbf{P}^1 \to B$ for the projection. We say $\mathcal{D}$ admits a *Tate twists*, if the object

$$\mathbf{1}_B(-1) := cof(\mathbf{1}_B \to f_*\mathbf{1}_{\mathbf{P}^1_B})$$

is $\otimes$-invertible. In that case, we will write

$$\mathbf{1}_B(1)$$

for its $\otimes$-inverse and call it the *Tate twist*.

Remark 1.2.5. Using proper base change, the Tate twist defines a cartesian section for $\mathcal{D}^*$. That is, we can define

$$\mathbf{1}_S(1)$$

for an arbitrary $S \in \mathcal{C}$, either by the above construction over S or via base change.

In the following, we will use the topology on $\mathcal{C}$ generated by families of cohomologically étale monomorphisms which jointly form a smooth cover (with respect to $\mathcal{D}$) and refer to it as the *open topology*. The following definition is taken from [Zav23][5.2.4].

Definition 1.2.6. We will say that $\mathcal{D}$ admits a *theory of first chern classes*, if it admits Tate twists, and it comes with a natural transformation

$$c_1\colon R\Gamma_{\mathrm{open}}(_, \mathbf{G}_{\mathrm{m}})[-1] \to \mathrm{Hom}(\mathbf{1}_{(_)}, \mathbf{1}_{(_)}(1))$$

of sheaves of spectra on $\mathcal{C}$.

Remark 1.2.7. Note that there is a natural transformation

$$R\Gamma_{\mathrm{Zar}}(_, \mathbf{G}_{\mathrm{m}})[-1] \to R\Gamma_{\mathrm{open}}(_, \mathbf{G}_{\mathrm{m}})[1]$$

of sheaves of spectra on $\mathrm{Sm}_B^{\mathrm{sep}}$. Furthermore, π_0 of the left-hand side computes the Picard group. In particular, any line bundle $\mathcal{L}$ on a smooth separated scheme S over B gives rise to a map

$$c_1(\mathcal{L})\colon \mathbf{1}_S \to \mathbf{1}_S(1).$$

In practice, π_0 of the right-hand side will essentially compute the group of line bundles on an object $S \in \mathcal{C}$.

Remark 1.2.8. The assignment of first Chern classes is compatible with base change and we have the formula

$$c_1(\mathcal{L}_1 \otimes \mathcal{L}_2) \simeq c_1(\mathcal{L}_1) + c_1(\mathcal{L}_2).$$

Construction 1.2.9 (The Projective bundle formula). Consider a morphism $f\colon X \to S$ of separated smooth schemes over B and a line bundle $\mathcal{L}$ on X. Then, by adjunction, the morphism $c_1(\mathcal{L})$ induces a morphism

$$c_1(\mathcal{L})\colon \mathbf{1}_S \to f_*\mathbf{1}_X(1)$$

which we will denote the same way. Furthermore, taking $\otimes$-powers iteratively of this morphism with itself, we obtain a map

$$c_1(\mathcal{L})^d\colon \mathbf{1}_S \to f_*\mathbf{1}_X(d).$$

[6]This is of course structure and not a property.

Now, if we specialize to $f\colon \mathbf{P}_S(\mathcal{F}) \to S$ being the projection from the projective space associated to a vector bundle $\mathcal{F}$ of rank $d+1$ on S, we can use these constructions to obtain a morphism

$$\sum_{i=0}^{d} c_1(\mathcal{O}(1))^i(d-i)\colon \bigoplus_{i=0}^{d} \mathbf{1}_S(d-i) \to f_*\mathbf{1}_{\mathbf{P}_S(\mathcal{F})}(d)$$

where $\mathcal{O}(1)$ denotes the universal line bundle and $c_1(\mathcal{O}(1))^0$ by adjunction corresponds to the identity.

The following definition is taken from [Zav23][5.2.8].

Definition 1.2.10. We will say that a theory of first Chern classes for $\mathcal{D}$ is an *additive orientation*, if for each $d \geq 1$ the map

$$\sum_{i=0}^{d} c_1(\mathcal{O}(1))^i(d-i)\colon \bigoplus_{i=0}^{d} \mathbf{1}_B(d-i) \to f_*\mathbf{1}_{\mathbf{P}^d_B}(d)$$

is an isomorphism.

Remark 1.2.11. Note that the Projective bundle formula isomorphism base changes to its counterpart over an arbitrary object $S \in \mathcal{C}$. In particular, the Projective Bundle formula holds over any object in $\mathcal{C}$ if we have an additive orientation. As we can check isomorphisms locally, we also obtain a Projective Bundle formula for an arbitrary projective bundle over a smooth separated scheme over S.

Remark 1.2.12. For any object $S \in \mathcal{C}$ and any sheaf $\mathcal{E} \in \mathcal{D}(S)$ we obtain an isomorphism

$$\sum_{i=0}^{d} c_1(\mathcal{O}(1))^i(d-i)\colon \bigoplus_{i=0}^{d} \mathcal{E}(d-i) \to f_*f^*\mathcal{E}(d)$$

by tensoring the sheaf on the Projective Bundle formula isomorphism.

1.3. **Smooth morphisms.** In this subsection, we will axiomatize what we want a class of smooth morphisms to satisfy in order to compute their dualizing sheaf. For this, we fix a geometric six-functor formalism $\mathcal{D}$.

Recall that we referred to the topology generated by families of cohomologically étale monomorphisms, which generated an étale covering, the *open topology*.

Definition 1.3.1. We will call a morphism $Z \to X$ in $\mathcal{C}$ a *regular immersion*, if it locally on the target in the open topology sits in a Cartesian square

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow \\ \{0\} & \longrightarrow & \mathbf{A}^1_B. \end{array}$$

Remark 1.3.2. We chose the wording regular immersion, as the above Cartesian square should be thought of as a derived Cartesian square.

Definition 1.3.3. Given two subcategories $(\text{Ét}, \text{Sm})$ of $\mathcal{C}$, we will refer to the morphisms in Ét as geometrically étale and to morphisms in Sm as geometrically smooth, if they satisfy the following:

(a) Both classes are stable under base change along arbitrary morphisms between objects in Sm and Ét $\subset$ Sm.
(b) Any morphism in Ét is cohomologically étale.
(c) $\mathrm{Sm}_B^{\mathrm{sep}} \subset \mathrm{Sm}$.
(d) Any morphism in Sm, locally on source and target in the open topology, factors as
$$X \xrightarrow{e} \mathbf{A}^n_S \xrightarrow{pr} S$$
where $e \in$ Ét and pr denotes the canonical projection.
(e) Given a morphism $f\colon X \to S$ in Sm, then any section $s\colon S \to X$ factors as
$$S \xrightarrow{i} U \xrightarrow{j} X.$$
where j is an étale monomorphism and i a regular immersion.
(f) For any commutative triangle
$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ & \searrow^{g} & \downarrow{\scriptstyle f} \\ & & S \end{array}$$
where g and f are geometrically smooth and i a regular immersion. Locally on X in the open topology, we can find cartesian squares
$$\begin{array}{ccccc} Z & = & Z & = & Z \\ {\scriptstyle i}\downarrow & & {\scriptstyle t}\downarrow & & \downarrow{\scriptstyle 0} \\ X & \leftarrow & U & \rightarrow & \mathbf{A}^n_Z \end{array}$$
with the horizontal morphisms étale.
(g) For any commutative triangle
$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ & \searrow^{g} & \downarrow{\scriptstyle f} \\ & & S \end{array}$$
where g and f are geometrically smooth and i is a regular immersion. There exists a geometrically smooth morphism $\mathbf{Bl}_Z(X) \to X$, which restricted to the open locus on X, where we have a cartesian square
$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow \\ \{0\} & \longrightarrow & \mathbf{A}^n_B \end{array}$$
is given by $X \times_{\mathbf{A}^n_B} \mathbf{Bl}_{\{0\}}(\mathbf{A}^n)$. Here we write $\mathbf{Bl}_{\{0\}}(\mathbf{A}^n_B)$ for the Blow-up from derived algebraic geometry [KR25] [Tan26][A].

Let us make some remarks on this definition.

Remark 1.3.4. Assertion (f) in the definition of geometrically smooth morphisms, in practice, follows from the Jacobian criterion together with the fact that the diagonal of an étale morphism is an open embedding and that the image of the complement of an open along a Zariski closed immersion has an open complement [MV99][Section 3. Lemma 2.28].

Remark 1.3.5. Given a Zariski closed immersion $Z \to X$ then locally on X, Z admits a complement by pulling back the complement of the zero section. As these are monomorphisms we can glue those local complements to a global complement $U \subset X$. Furthermore for any etale morphism $V \to X$, such that the square

$$\begin{array}{ccc} Z & \longrightarrow & V \\ \| & & \downarrow \\ Z & \longrightarrow & X \end{array}$$

is Cartesian, we see that the pair

$$V \amalg U \to X$$

is a cohomologically étale covering, since this holds locally. Note also that as $U \subset X$ is a monomorphism, satisfying descent for this dét cover, is equivalent to sending the square

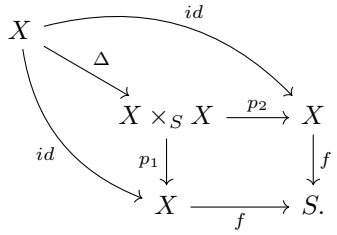
$$\begin{array}{ccc} U \times_X V & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & X \end{array}$$

to a pullback.

Remark 1.3.6. Condition (h) is a property.

In order to prove the cohomologically smoothness of a geometrically smooth morphism we will use the following criterion from [Zav23][3.2.4.].

Consider a morphism $f\colon X \to S$ in $\mathcal{C}_E$. Then we consider the commutative diagram



Definition 1.3.7. A *trace-cycle theory* on a morphism $f\colon X \to S$ in $\mathcal{C}_E$ consists of a triple $(\omega_f, \mathrm{tr}_f, \mathrm{cl}_\Delta)$ of:

(a) A $\otimes$-invertible object ω_f in $\mathcal{D}(X)$.
(b) A *trace morphism* $\mathrm{tr}_f\colon f_!\omega_f \to \mathbf{1}_S$ in $\mathcal{D}(S)$.
(c) A *cycle* morphism $\mathrm{cl}_\Delta\colon \Delta_!\mathbf{1}_X \to (p_2)^*\omega_f$ in $\mathcal{D}(X \times_S X)$.

Such that the following hold:

(1) The composition

$$\mathbf{1}_X \xrightarrow{\simeq} (p_1)_!\Delta_!\mathbf{1}_X \xrightarrow{(p_1)_!(\mathrm{cl}_\Delta)} (p_1)_!(p_2)^*\omega_f \xrightarrow{\mathrm{tr}_{p_1}} \mathbf{1}_X$$

is the identity. Where we write $\mathrm{tr}_{p_1} \simeq f^*(\mathrm{tr}_f)$.

(2) The composition

$$\omega_f \xrightarrow{\simeq} (p_2)_!(p_1^*\omega_f \otimes \Delta_!\mathbf{1}_X) \xrightarrow{(p_2)_!(id\otimes \mathrm{cl}_\Delta)} (p_2)_!(p_1^*\omega_f \otimes p_2^*\omega_f) \simeq (p_2)_!p_2^*\omega_f \otimes \omega_f \xrightarrow{\mathrm{tr}_f\otimes id} id \otimes \omega_f \simeq \omega_f$$

is the identity.

Remark 1.3.8. The definition unwinds what it means to have a unit and counit of an adjunction

$$\mathbf{1}_X \dashv \omega_f$$

in $\mathcal{K}_S$.

Then section 3 in [Zav23] implies the following.

Theorem 1.3.9 (**Zavyalov**). *Consider a geometrized geometric setup $(\mathcal{C}, \mathcal{C}_E)$ and a geometric six-functor formalism $\mathcal{D}$ on it. Then a morphism $f\colon X \to S$ in $\mathcal{C}_E$ is cohomologically smooth, if and only if it admits a trace-cycle theory, and in that case, we can compute the dualizing sheaf as*

$$f^! \mathbf{1}_S \simeq \omega_f.$$

Furthermore, for any class of geometrically smooth morphisms, the following are equivalent:

(a) Any geometrically smooth morphism is cohomologically smooth.
(b) The morphism $\mathbf{A}^1 \to B$ is cohomologically smooth.
(c) The morphism $\mathbf{P}^1 \to B$ is cohomologically smooth.

Proof. Using conditions (b) and (d) this follows as in [Zav23][3.3.3.]. □

1.4. **Vector bundles and Thom spaces.** In this section, we will define the Thom space associated with a vector bundle and compute its cohomology. For this, we fix an additively oriented six-functor formalism $\mathcal{D}$ on $\mathcal{C}$ and recall that we have a functor

$$\mathrm{Sm}_B^{\mathrm{sep}} \to \mathcal{C}$$

which preserves those finite limits, which are also finite limits in the category of all schemes.

1.4.1. For a morphism $f\colon X \to S$ in $\mathcal{C}$ and an object $\mathcal{E}_S \in \mathcal{D}(S)$, we will write

$$\mathbf{C}^*(X, \mathcal{E}_S) := f_* f^* \mathcal{E}_S \in \mathcal{D}(S)$$

for its cohomology. Furthermore if f lives in $\mathcal{C}_E$, we will write

$$\mathbf{C}_*(X, \mathcal{E}_S) := f_! f^! \mathcal{E}_S \in \mathcal{D}(S)$$

for its homology. We will omit the coefficients in case $\mathcal{E}_S \simeq \mathbf{1}_S$.

We now also fix a class of geometrically smooth and geometrically étale morphisms.

1.4.2. Given an object $S \in \mathcal{C}$, we will write

$$(\mathrm{Sm}_S^{\mathrm{geom}})_{\mathrm{open}}$$

for the topos generated by geometrically smooth objects over S equipped with the topology generated by families of cohomologically étale monomorphisms, which jointly induce a $\mathcal{D}^*$-covering. Clearly, cohomology and homology induce functors on this topos.

We can now consider the following constructions.

1.4.3 (**Vector bundles**)**.** Given an object $S \in \mathcal{C}$, we will write

$$\mathrm{B}\mathbf{Gl}_{\mathrm{n}S} \in (\mathrm{Sm}_S^{\mathrm{geom}})_{\mathrm{open}}$$

for the stack of rank n vector bundles on S and

$$\mathbf{Vect}_S := \coprod_n \mathrm{B}\mathbf{Gl}_{\mathrm{n}S} \in (\mathrm{Sm}_S^{\mathrm{geom}})_{\mathrm{open}}$$

for the stack of all vector bundles on S.

That is a map $\langle \mathcal{F} \rangle \colon X \to \mathrm{B}\mathbf{Gl}_{\mathrm{n}S}$ corresponds to a $\mathbf{Gl}_{\mathrm{n}}$-torsor

$$\mathbf{V}_X(\mathcal{F}) \to X$$

which trivializes locally in the open topology.

1.4.4 (**Projective bundles**)**.** Given a rank ≥ 1 vector bundle $\mathbf{V}_X(\mathcal{F}) \to X$, we have a zero section $0 \colon X \to \mathbf{V}_X(\mathcal{F})$. Furthermore using the diagonal map $\mathbf{G}_{\mathrm{m}} \to \mathbf{Gl}_{\mathrm{n}}$, we obtain an $\mathbf{G}_{\mathrm{m}X}$-action on $\mathbf{V}_X(\mathcal{F})$, which restricts to $\mathbf{V}_X(\mathcal{F}) - \{0\}$.

From this we define the *projective bundle* associated to $\mathbf{V}_X(\mathcal{F})$ to be

$$\mathbf{P}_X(\mathcal{F}) := (\mathbf{V}_X(\mathcal{F}) - \{0\})/\,\mathbf{G}_{\mathrm{m}} \in (\mathrm{Sm}_S^{\mathrm{geom}})_{\mathrm{open}}.$$

Note that there is a canonical map $\mathbf{P}_X(\mathcal{F}) \to \mathrm{B}\mathbf{G}_{\mathrm{m}}$, which defines a line bundle $\mathcal{O}(1)$ on the projective bundle. This line bundle has a first Chern class.

Remark 1.4.5. Any projective bundle is weakly cohomologically proper, as locally on the base, this is true by assumption.

1.4.6 (**Thom spaces**)**.** Given a rank ≥ 1 vector bundle $\mathbf{V}_X(\mathcal{F})$, we define the its *Thom space* to be the cofibre

$$\mathbf{Th}_X(\mathcal{F}) := cof(\mathbf{P}_X(\mathcal{F}) \to \mathbf{P}_X(\mathcal{F} \oplus \mathcal{O}_X)) \in (\mathrm{Sm}_S^{\mathrm{geom}})_{\mathrm{open}}.$$

We now have the following classical observation.

Proposition 1.4.7. *Given an object $S \in \mathcal{C}$ together with a vector bundle $\mathbf{V}_S(\mathcal{F}) \to S$ of rank $n \geq 1$. Then we have a canonical isomorphism*

$$t(\mathcal{F}) \colon \mathbf{1}_S(-n) \xrightarrow{\sim} \mathbf{C}^*(\mathbf{Th}_S(\mathcal{F}))$$

in $\mathcal{D}(S)$. In particular, the cohomology of those Thom spaces is $\otimes$-invertible.

Proof. See for example [AHI25][6]. We consider the following diagram, which commutes by naturality of first Chern classes.

$$\begin{array}{ccc} \mathbf{C}^*(\mathbf{P}_S(\mathcal{F} \oplus \mathcal{O}_S)) & \longrightarrow & \mathbf{C}^*(\mathbf{P}_S(\mathcal{F})) \\ \simeq\uparrow & & \uparrow\simeq \\ \bigoplus_{i=0}^{n} \mathbf{1}_S(-i) & \longleftarrow & \bigoplus_{i=0}^{n-1} \mathbf{1}_S(-i). \end{array}$$

Here, the vertical isomorphisms come from the Projective bundle formula. Thus, we obtain an isomorphism between the cofibre of the lower horizontal and the fiber of the upper horizontal map. □

1.5. **Tang's construction of cycle classes.** To construct a cycle class theory for a geometrically smooth morphism, we will make use of a construction of Longke Tang [Tan26].

Let us now fix a geometric six-functor formalism $\mathcal{D}$, which comes equipped with an additive orientation. Furthermore, we fix a class of geometrically smooth and geometrically étale morphisms.

1.5.1. For $S \in \mathrm{Sm}$ we write $\mathrm{Sm}_S^{\mathrm{geom}}$ for the category of geometrically smooth morphisms over S. Furthermore, we will call a blow-up square

$$\begin{array}{ccc} E & \longrightarrow & \mathbf{Bl}_Z(X) \\ \downarrow & \overset{f}{\searrow} & \downarrow{\scriptstyle p} \\ Z & \xrightarrow[i]{} & X \end{array}$$

associated to a regular closed immersion $i\colon Z \to X$ between objects in $\mathrm{Sm}_S^{\mathrm{geom}}$ a *smooth blowup square*.

We will also use the following minor assumption on our six-functor formalism.

Definition 1.5.2. We will say that $\mathcal{D}$-cohomology satisfies *blow-up excision* if for every $n \geq 1$ the blow-up square

$$\begin{array}{ccc} \mathbf{P}_B^{n-1} & \longrightarrow & \mathbf{V}_{\mathbf{P}_B^{n-1}}(\mathcal{O}(1)) \\ \downarrow & \overset{f}{\searrow} & \downarrow{\scriptstyle p} \\ B & \xrightarrow[0]{} & \mathbf{A}_B^n \end{array}$$

induces a (co)cartesian square

$$\begin{array}{ccc} \mathbf{1}_{\mathbf{A}_B^n} & \longrightarrow & 0_*0^*\mathbf{1}_{\mathbf{A}_B^n} \\ \downarrow & & \downarrow \\ p_*p^*\mathbf{1}_{\mathbf{A}_B^n} & \longrightarrow & f_*f^*\mathbf{1}_{\mathbf{A}_B^n} \end{array}$$

in $\mathcal{D}(\mathbf{A}_B^n)$.

From this definition, we can obtain the following general form of blow-up excision.

Proposition 1.5.3. *Consider an object $S \in Sm$. Then any blow-up square in Sm_S^{geom} as in 1.5.1 induces a (co)cartesian square*

$$\begin{array}{ccc} \mathbf{1}_X & \longrightarrow & i_*i^*\mathbf{1}_X \\ \downarrow & & \downarrow \\ p_*p^*\mathbf{1}_X & \longrightarrow & f_*f^*\mathbf{1}_X \end{array}$$

in $\mathcal{D}(X)$.

PROOF. By pulling back along the structure map (using proper base change), we can deduce the assertion for blow-up squares of the form

$$\begin{array}{ccc} \mathbf{P}^{n-1}_Z & \longrightarrow & \mathbf{V}_{\mathbf{P}^{n-1}_Z}(\mathcal{O}(1)) \\ \downarrow & \overset{f}{\searrow} & \downarrow{\scriptstyle p} \\ Z & \xrightarrow[0]{} & \mathbf{A}^n_Z \end{array}$$

for general $Z \in \mathrm{Sm}$. For a general smooth blow-up square, we can use assertion (f) in the definition of geometrically smooth morphisms to Zariski locally on X find cartesian squares

$$\begin{array}{ccccc} Z & \longleftarrow & Z & \longrightarrow & Z \\ {\scriptstyle i}\downarrow & & \downarrow & & \downarrow{\scriptstyle 0} \\ X & \xleftarrow[j]{} & U & \xrightarrow[e]{} & \mathbf{A}^n_Z. \end{array}$$

Let us write Q_X, Q_U and $Q_{\mathbf{A}^n_Z}$ for the cohomology squares induced by the corresponding blow-up squares. Then by open descent, to check that Q_X is cartesian, it suffices to see that

$$j^*(Q_X) \simeq Q_U \simeq e^*(Q_{\mathbf{A}^n_Z})$$

is cartesian. But this we did see in the first half of the argument. □

Remark 1.5.4. If $\mathcal{D}$-cohomology over S satisfies blow-up excision, by taking duals, one sees that also the square

$$\begin{array}{ccc} f_! f^! \mathbf{1}_X & \longrightarrow & p_! p^! \mathbf{1}_X \\ \downarrow & & \downarrow \\ i_! i^! \mathbf{1}_X & \longrightarrow & \mathbf{1}_X \end{array}$$

associated to a blow-up square is (co)cartesian in $\mathcal{D}(X)$ (note that all maps appearing are weakly cohomologically proper).

Remark 1.5.5. The author does not know how to formally deduce blowup-excision from the projective bundle formula. In practice, though, it will normally be possible to make a reduction of this type. Essentially, because the following is true.

Proposition 1.5.6. *For any $S \in Sm$ and any object $\mathcal{E}_S \in \mathcal{D}(S)$, the functor*

$$\mathbf{C}_*(_, \mathcal{E}_S) \colon Sm_S^{geom} \to \mathcal{D}(S)$$

as well as the functor

$$\mathbf{C}^*(_, \mathcal{E}_S) \colon (Sm_S^{geom})^{op} \to \mathcal{D}(S)$$

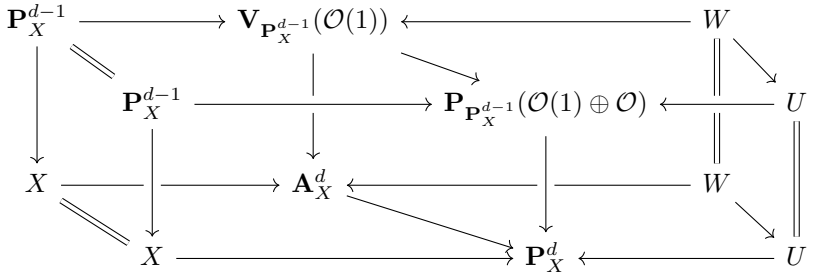
sent smooth blow-up squares to pushout and pullback squares.

PROOF. We first prove the claim for cohomology. As we assume that cohomology is a Nisnevich sheaf, by [AHI25][2.2] it suffices to check that for any geometrically smooth X a Blow-up squares of the form

$$\begin{array}{ccc} \mathbf{P}^{d-1}_X & \longrightarrow & \mathbf{V}_{\mathbf{P}^{d-1}_X}(\mathcal{O}(1)) \\ \downarrow & & \downarrow \\ X & \xrightarrow[0]{} & \mathbf{A}^d_X \end{array}$$

get send to (co)cartesian squares by $\mathcal{E}_S$-cohomology (here we are using assertion (f) in the definition of geometrically smooth morphisms).

Now consider the commutative diagram



where we write U and W for the respective complements of the zero sections. Now, by Zariski descent, the upper and lower squares in the right cube get send to (co)cartesian squares by $\mathcal{E}_S$-cohomology. From this one sees that on $\mathcal{E}_S$-cohomology the square in the back of the left cube becomes (co)cartesian if and only if the square in the front of the left cube becomes (co)cartesian.

Thus, it suffices to check the claim for squares of the form

$$\begin{array}{ccc} \mathbf{P}^{d-1}_X & \longrightarrow & \mathbf{P}_{\mathbf{P}^{d-1}_X}(\mathcal{O}(1)\oplus\mathcal{O}) \\ \downarrow & \overset{h}{\searrow} & \downarrow f \\ X & \xrightarrow[0]{} & \mathbf{P}^d_X. \end{array}$$

For the proof, let us refer to such squares as a projective Blow-up squares. Then we first claim the following:

$(*)$ The proposition holds for projective Blow-up squares with $X = S$.

Let us write $p\colon \mathbf{P}^d_S \to S$ for the projection. Then we have to check that the square

$$\begin{array}{ccc} p_*p^*\mathcal{E}_S & \longrightarrow & p_*0_*0^*p^*\mathcal{E}_S \\ \downarrow & & \downarrow \\ p_*f_*f^*p^*\mathcal{E}_S & \longrightarrow & p_*h_*h^*p^*\mathcal{E}_S \end{array}$$

is (co)cartesian. Using the Projective Bundle formula, we see that this square identifies with the square

$$\begin{array}{ccc} \oplus^d_{i=0}\mathcal{E}_S(-i) & \longrightarrow & \mathcal{E}_S \\ \downarrow & & \downarrow \\ \oplus^{d-1}_{i=0}\mathcal{E}_S(-i)\oplus\oplus^d_{i=1}\mathcal{E}_S(-i) & \longrightarrow & \oplus^{d-1}_{i=0}\mathcal{E}_S(-i) \end{array}$$

which is easily seen to be (co)cartesian. To see that the map induced on the fibers is the identity one uses that

$$f^*\mathcal{O}(1) \simeq \mathcal{O}(1).$$

Now, for a general projective Blow-up square, let us write $g\colon X \to S$ for the structure map. Then applying $*$ for $X = S$ with coefficients in $g^*\mathcal{E}_S$ and using that g_*

preserves (co)cartesian squares, we win. This finishes the argument for cohomology. To see the claim for homology, we reduce the claim to projective Blow-up squares over the base in the same way, and for those, one uses that for a proper map $f\colon X \to S$, we have an identification

$$f_! f^! \mathcal{E}_S \simeq \underline{\mathrm{Hom}}_S(f_* \mathbf{1}_X, \mathcal{E}_S)$$

such that the claim follows from the case of cohomology applied to the unit. □

In the following, we will use the notion of deflatability, which first appeared in [CTHK97] (see also [BK26]). For this, we consider the following commutative diagram, which we can associate with an object $S \in \mathcal{C}$.

$$\begin{array}{ccccc} \mathbf{A}^1_S & \xrightarrow{j} & \mathbf{P}^1_S & \xleftarrow{0} & S \\ & \searrow^{\pi} & \downarrow{\scriptstyle \overline{\pi}} & \swarrow & \\ & & S. & & \end{array}$$

Proposition 1.5.7. *Given an object $S \in \mathcal{C}$, we have an identification of the morphisms*

$$\mathbf{C}^*(\mathbf{P}^1_S) \underset{(\pi\circ 0)^*}{\overset{j^*}{\rightrightarrows}} \mathbf{C}^*(\mathbf{A}^1_S)$$

in $\mathcal{D}(S)$. This construction is functorial in S.

Proof. We follow the proof given in [CTHK97][5.4.3]. That is, we consider the following diagram

$$\begin{array}{ccc} \mathbf{C}^*(\mathbf{P}^1_S) & \xrightarrow{j^*} & \mathbf{C}^*(\mathbf{A}^1_S) \\ \simeq\uparrow & \searrow^{0^*} & \uparrow{\scriptstyle \pi^*} \\ \mathbf{C}^*(S) \oplus \mathbf{C}^*(S)(-1) & & \mathbf{C}^*(S) \end{array}$$

where the left vertical isomorphism comes from the projective bundle formula. We want to see that the triangle on the right commutes. To check this, we can precompose with the just explained isomorphism. On the factor to the left, this follows from the functoriality of cohomology, and on the factor on the right, both compositions are given by the first Chern class of $\mathcal{O}_{\mathbf{A}^1_S}$. □

We are now ready to recall the construction of cycle class maps from [Tan26].

1.5.8 (Cycle class map). We follow the construction given in [Tan26][3.1]. Consider a regular closed immersion $i\colon Z \to X$ between two geometrically smooth objects over S. Then we can produce the following blowup squares

$$\begin{array}{ccc} \mathbf{P}_Z(\mathcal{N}_i) & \longrightarrow & \mathbf{Bl}_Z(X) \\ \downarrow & & \downarrow \\ Z & \longrightarrow & X \end{array} \qquad \begin{array}{ccc} \mathbf{P}_Z(\mathcal{N} \oplus \mathcal{O}) & \longrightarrow & \mathbf{Bl}_Z(X \times \mathbf{P}^1) \\ \downarrow & & \downarrow \\ Z \times \{0\} & \longrightarrow & X \times \mathbf{P}^1. \end{array}$$

Note that there is a canonical map of squares from the left to the right. As we assume blow-up excision 1.5.3, this map induces a map between (co)cartesian squares

$$\begin{array}{ccc} \mathbf{C}^*(\mathbf{P}^1_X) & \longrightarrow & \mathbf{C}^*(Z) \\ \downarrow & & \downarrow \\ \mathbf{C}^*(\mathbf{Bl}_Z(\mathbf{P}^1_X)) & \longrightarrow & \mathbf{C}^*(\mathbf{P}^1_Z(\mathcal{N}_i \oplus \mathcal{O})) \end{array} \qquad \begin{array}{ccc} \mathbf{1}_X & \longrightarrow & \mathbf{C}^*(Z) \\ \downarrow & & \downarrow \\ \mathbf{C}^*(\mathbf{Bl}_Z(X)) & \longrightarrow & \mathbf{C}^*(\mathbf{P}_Z(\mathcal{N}_i)) \end{array}$$

in $\mathcal{D}(X)$. Thus, taking fibers at each term of this map, we obtain a (co)fiber sequence

$$\mathbf{C}^*(\mathbf{P}^1_X/X \times \{0\}) \to Q \to \mathbf{C}^*(\mathbf{Th}_Z(\mathcal{N}_1))$$

. Furthermore the one section $X \times \{1\} \to \mathbf{Bl}_Z(\mathbf{P}^1_X)$ induces a map

$$Q \to \mathbf{1}_X$$

who, using 1.5.7, identifies with the zero map after precomposing to $\mathbf{C}^*(\mathbf{P}^1_X/X \times \{0\})$. This induces a map

$$\mathbf{C}^*(\mathbf{Th}_Z(\mathcal{N}_i)) \to \mathbf{1}_X$$

and using 1.4.7, we obtain a map

$$\mathrm{cl}_i \colon i_*\mathbf{1}_Z \to \mathbf{1}_X(n)$$

in $\mathcal{D}(X)$, where n is the codimension of Z in X.

Definition 1.5.9. For a regular closed immersion $i \colon Z \to X$ between two objects in $\mathrm{Sm}^{\mathrm{geom}}_S$, we will refer to the map

$$\mathrm{cl}_i \colon i_*\mathbf{1}_Z \to \mathbf{1}_X(n)$$

constructed in 1.5.8 as the *cycle class map* associated to i.

The rest of this section is devoted to proving the following. The statement asserts that on $\mathrm{Sm}^{\mathrm{sep}}_B$ our theory of first Chern classes underlies a theory of cycle classes in the sense of [Zav23][5.3.3.]. The cycle classes we use are the ones constructed above in the case that the regular immersion is given by an effective Cartier divisor.

Theorem 1.5.10. *Consider an effective cartier divisor $i \colon D \to S$ in Sm^{sep}_B. Then the composition*

$$cl_i \circ ad^*_* \colon \mathbf{1}_S \to i_*\mathbf{1}_Z \to \mathbf{1}_S(1)$$

canonically identifies with the first Chern class associated to the line bundle $\mathcal{O}(D)$.

We will start by recalling some computations (mainly taken from [Tan26] and [Tan]).

1.5.11 (Cohomology of $\mathbf{P}^\infty$). Consider the canonical diagram

$$\begin{array}{ccccccccc} S & \longrightarrow & \mathbf{P}^1_S & \longrightarrow & \mathbf{P}^2_S & \longrightarrow & \mathbf{P}^3_S & \longrightarrow & \dots \\ & \searrow & & \overset{p_1}{\searrow} & \downarrow{\scriptstyle p_2} & \swarrow{\scriptstyle p_3} & & & \\ & & & & S & & & & \end{array}$$

over some separated smooth scheme S over B and let us write $\mathbf{P}^\infty_S$ for the colimit. We can compute the cohomology of $\mathbf{P}^\infty_S$ by the formula

$$\lim_n (p_n)_*(p_n)^*\mathbf{1}_S \simeq \prod_{i=0}^{\infty} \mathbf{1}_S(-i) \in \mathcal{D}(S).$$

By 1.5.7, we see that cohomology identifies $\mathbf{P}^1$-homotopic maps and thus also (non-trivially) weighted $\mathbf{A}^1$-homotopic maps (see [AHI25][4.6.]). We will use this in the following computations.

1.5.12. An example of a weighted homotopy equivalence is the map

$$V/\,\mathrm{B}\mathbf{G}_\mathrm{m} \to \mathrm{B}\mathbf{G}_\mathrm{m}$$

where V denotes a vector bundle over a smooth and separated B-scheme and $\mathbf{G}_\mathrm{m}$ acts on V with non-zero weight. From this, one also sees that the map

$$cof((V/\,\mathrm{B}\mathbf{G}_\mathrm{m}) - (\mathrm{B}\mathbf{G}_\mathrm{m}) \to V/\,\mathrm{B}\mathbf{G}_\mathrm{m}) \to cof(\mathcal{O}(1)\colon \mathbf{P}(V) \to \mathrm{B}\mathbf{G}_\mathrm{m})$$

is a weighted homotopy equivalence. In particular, we see that we obtain an isomorphism

$$\mathbf{C}^*(\mathrm{B}\mathbf{G}_\mathrm{m}\,/S) \xrightarrow{\sim} \mathbf{C}^*((V/\,\mathbf{G}_\mathrm{m})/S)$$

(and similar for the second map) in $\mathcal{D}(S)$.

1.5.13 (**Cohomology of** $\mathrm{B}\mathbf{G}_\mathrm{m}$)**.** Consider a vector bundle $\mathcal{F}$ on S for which we can find a surjective map $\mathcal{F} \to \mathcal{O}$. Then using $\mathbf{P}^1$-homotopies 1.5.7 and elementary blow up excision 1.5.6 one proves the same way as in [AHI25][5.3] that the projection

$$\mathbf{C}^*(\mathrm{B}\mathbf{Gl}_\mathrm{n}\,/S) \xrightarrow{\sim} \mathbf{C}^*(\mathbf{Gr}_\mathrm{n}(\mathcal{F}^\infty)/S)$$

from the infinite Grassmannian is an isomorphism in $\mathcal{D}(S)$.

A special case of this computation is that the map

$$\mathbf{P}^\infty \to \mathrm{B}\mathbf{G}_\mathrm{m}$$

induces an isomorphism on cohomology. In particular, using 1.5.11 we see that we have a canonical identification

$$\mathbf{C}^*(\mathrm{B}\mathbf{G}_\mathrm{m}\,/S) \simeq \prod_{i=0}^{\infty} \mathbf{1}_S(-i)$$

in $\mathcal{D}(S)$.

1.5.14 (**Description of first chern classes**)**.** Using 1.5.13 one sees that given a line bundle $\mathcal{L}$ over S corresponding to a map

$$i_\mathcal{L}\colon S \to \mathrm{B}\mathbf{G}_\mathrm{m}\,.$$

The first Chern class of $\mathcal{L}$ is given by the composition

$$\mathbf{1}_S \to \prod_{i=0}^{\infty} \mathbf{1}_S(1-i) \simeq \mathbf{C}^*(\mathrm{B}\mathbf{G}_\mathrm{m}\,/S)(1) \to \mathbf{1}_S(1)$$

where the first map is the inclusion into the first factor, and the second map is induced by $i_\mathcal{L}$ on cohomology.

1.5.15 (**Normalisation of cycle class maps**)**.** Using $\mathbf{P}^1$-homotopies 1.5.7 and elementary blow up excision 1.5.6, as in the proof of [Tan26][3.10], on sees the following. Given a vector bundle V over a smooth separated B-scheme, the cycle class map of the zero section $S \to V$ is refined by a map

$$\mathbf{C}^*(\mathbf{Th}_S(V)/V) \to \mathbf{C}^*(V/V - \{0\}/V)$$

which admits a section. This section is functorial in S and V (see [Tan26][2.17]).

Proof of 1.5.10. Note first that the assertion is stable under base change, so it suffices to check the claim for the universal effective Cartier divisor

$$\begin{array}{ccc} G = \mathrm{B}\mathbf{G}_{\mathrm{m}} & \xrightarrow{0} & V = \mathbf{A}^1/\mathbf{G}_{\mathrm{m}} \\ & \searrow^{id} & \downarrow p \\ & & G = \mathrm{B}\mathbf{G}_{\mathrm{m}}. \end{array}$$

That is, we have to check that the composition

$$\mathrm{cl}_0 \circ \mathrm{ad}^*_*\colon \mathbf{1}_V \to 0_*\mathbf{1}_G \to \mathbf{1}_V(1)$$

identifies with the first Chern class associated to the line bundle $\mathcal{L}^{\mathrm{univ}}$ corresponding to the map

$$p\colon V \to G.$$

By adjunction, this map corresponds to the map

$$\mathbf{1}_S(-1) \to \mathbf{C}^*(\mathbf{Th}_G(\mathcal{L}^{\mathrm{univ}}_{|G})/S) \to \mathbf{C}^*(V/S)$$

in $\mathcal{D}(S)$, where the first map comes from the structure morphism and the second from 1.5.8. As we are in the situation of 1.5.15, we see that the second map of this composition identifies with the map

$$\mathbf{C}^*(\mathbf{Th}_G(\mathcal{L}^{\mathrm{univ}}_{|G})/S) \to \mathbf{C}^*(V/V - \{0\}/S) \to \mathbf{C}^*(V/S)$$

where the first map is the section from 1.5.15 and the second map is induced by the canonical projection. Finally, using the identification from 1.5.12 and 1.5.13 we see that the map in question can be described as the composition

$$\mathbf{1}_S(-1) \to \prod_{i=1}^{\infty} \simeq \mathbf{C}^*(G/\mathbf{P}^1/S) \to \mathbf{C}^*(V/S)$$

where the first map is the inclusion into the first factor and the second map is induced by p. This identifies with the first Chern class of $\mathcal{L}^{\mathrm{univ}}$ by 1.5.14. □

1.6. **Cohomological smoothness.** In this section, we will argue that any geometrically smooth morphism will be cohomologically smooth through the eyes of our six-functor formalism. We will also compute the dualizing sheaf of such a morphism.

For this, we fix an additively oriented six-functor formalism $\mathcal{D}$ on $\mathcal{C}$, for which cohomology satisfies elementary blow-up excision. Furthermore, we fix a class of geometrically smooth and geometrically étale morphisms.

We start with the construction of a trace morphism for the morphism $f\colon \mathbf{P}^1_B \to B$. For this, we follow [Zav23][5.6.1].

Construction 1.6.1. As a dualizing sheaf, we want to consider the Tate twist. That is, we have to construct a morphism

$$\mathrm{tr}_f\colon f_*\mathbf{1}_{\mathbf{P}^1_B}(1) \to \mathbf{1}_B.$$

Such a morphism is given by the composition

$$f_*\mathbf{1}_{\mathbf{P}^1_B}(1) \to \mathbf{1}_B \oplus \mathbf{1}_B(1) \to \mathbf{1}_B.$$

The first map is the inverse of the Projective Bundle formula isomorphism, and the second is the projection to the first factor.

Using the work [Zav23], we obtain the following.

Corollary 1.6.2. *Any geometrically smooth morphism $f\colon X \to S$ is cohomologically smooth for $\mathcal{D}$. Furthermore, if the morphism factors as*

$$f = p \circ e\colon X \to \mathbf{A}^n_S \to S$$

where e is cohomologically étale and p the projection, we have an identification

$$f^!(\mathbf{1}_S) \simeq \mathbf{1}_X(n)$$

in $\mathcal{D}(X)$.

Proof. By 1.3.9, to deduce the first claim, it suffices to check the following:

(∗) The triple $(\mathbf{1}_{\mathbf{P}^1_B}(1), \mathrm{tr}_f, \mathrm{cl}_\Delta)$ constructed in 1.6.1 and 1.5.8 gives a trace-cycle theory for the projection $f\colon \mathbf{P}^1_B \to B$.

But by 1.5.10 we see that our theory of Chern classes underlie our theory of cycle classes, allowing us to apply [Zav23][5.6.6]. The computation of the dualizing sheaf now easily follows by base change and the fact that for a cohomologically étale morphism $f\colon U \to S$, we have

$$f^* \simeq f^!.$$

□

Recall the category of motivic spectra $\mathrm{MS}(B)$ from [AHI25]. Then we also obtain the following corollary, which allows us to use the assertions from [Tan26].

Corollary 1.6.3. *Associating to a smooth separated B-scheme $f\colon X \to B$ its homology*

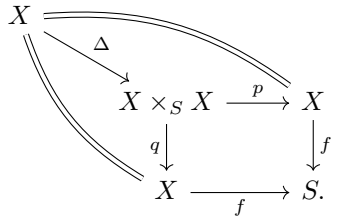
$$\mathbf{C}_*(X/B) \in \mathcal{D}(B)$$

induces a symmetric monoidal functor

$$MS(B) \to \mathcal{D}(B).$$

Proof. As any such morphism $f\colon X \to B$ is cohomologically smooth for $\mathcal{D}$ by 1.6.2, taking homology is symmetric monoidal. So the assertion follows from the universal property of $\mathrm{MS}(B)$ using 1.5.6 and 1.4.7. □

Construction 1.6.4. Let us consider a geometrically smooth morphism $f\colon X \to S$, then we can produce the following diagram:



Furthermore by assertion (e) in 1.3.3, there is a factorization

$$\Delta \simeq j \circ \tilde{\Delta}\colon X \to U \to X \times_S X$$

into a Zariski closed immersion followed by an open immersion. Thus, using 1.5.8, we obtain a cycle class map

$$\mathrm{cl}_{\tilde{\Delta}}\colon \tilde{\Delta}_!\mathbf{1}_X \to \mathbf{1}_U(d).$$

Furthermore, applying $j_!$ and using the counit, we obtain a map

$$\mathrm{cl}_\Delta \colon \Delta_! \mathbf{1}_X \to \mathbf{1}_{X \times_S X}(d)$$

which we will also refer to as *cycle class map*.

We now obtain the following theorem.

Theorem 1.6.5. *Given a geometrically smooth morphism $f \colon X \to S$, then the data*

(a) $\boldsymbol{1}_X(d)$
(b) $cl_\Delta \colon \Delta_! \boldsymbol{1}_X \to \boldsymbol{1}_{X \times_S X}(d)$

is part of a trace-cycle theory. Where d denotes the relative dimension of f.

Proof. We claim that the cycle class map is the unit of an adjunction $\mathbf{1}_X \dashv \mathbf{1}_X(d)$. Note that the existence of a counit for this adjunction is a property. Namely the property, that for any $T \to S \in (\mathcal{C}_E)_{/S}$ and any two objects $\mathcal{E} \in \mathcal{D}(T \times_S X)$ and $\mathcal{F} \in \mathcal{D}(T)$ the canonical map

$$\mathrm{Hom}_{\mathcal{D}(T)}(p_!\mathcal{E}, \mathcal{F}) \to \mathrm{Hom}_{\mathcal{D}(T \times_S X)}(\mathcal{E}, p^*\mathcal{F}(d))$$

induced by the cycle class map is an isomorphism (see [Lur, Tag 02CU]). Here we write $p \colon T \times_S X \to T$ for the projection. This can be checked cohomologically étale locally on X and S.

So by 1.3.3(d) we can assume that f factors as an étale map followed by a projection from an affine space. As a cycle class theory can be pulled back along a cohomologically étale morphism on the domain, we can further assume that f is given by a projection from an affine space and as trace-cycle theories are stable under base change, we can even assume that f is given by the map $\mathbf{A}^n_B \to B$.

Now note that a regular closed immersion locally factors as a composition of effective Cartier divisors. So using the compatibility of cycle class maps with composition [Tan26][3.9, 3.17] (here we make use of 1.6.3) and again stability of base change of trace-cycle theories, we can assume f is given by the projection $\mathbf{A}^1_B \to B$. Now it suffices to construct a trace-cycle theory for the map $\mathbf{P}^1_B \to B$, which we did in 1.6.2. □

Corollary 1.6.6. *For any geometrically smooth morphism $f \colon X \to S$, we obtain a canonical identification*

$$f^!(\boldsymbol{1}_S) \simeq \boldsymbol{1}_X(d)$$

in $\mathcal{D}(X)$, where d denotes the relative dimension of f.

2. Formal schemes as analytic stacks

2.1. **Recollections on analytic stacks.** In this text, we will make use of the theory of analytic stacks introduced by Clausen and Scholze [CS24]. We will now recall the examples of those we will use.

2.1.1. Recall that an affine analytic stack is represented by a pair

$$(A, \mathcal{D}(A))$$

where A is a (light) condensed ring and $\mathcal{D}(A)$ a full subcategory, stable under limits and colimits, of the category of modules over A in (derived) condensed abelian

groups. This subcategory is such that it inherits a closed symmetric monoidal structure and a t-structure from those A-modules.

Example 2.1.2. We will write $\mathrm{Spa}(\mathbf{Z})$ for the analytic stack represented by the pair $(\mathbf{Z}, \mathrm{Solid}(\mathbf{Z}))$ where the subcategory is given by solid abelian groups [CS24][Lecture 5] [CS19]. All our analytic stacks will live over $\mathrm{Spa}(\mathbf{Z})$.

Example 2.1.3. An important object is given by the following construction. Consider the profinite set

$$\mathbf{N} \cup \{\infty\} := \lim_n \{0, \dots, n, \infty\}$$

where the transition maps send the highest number to ∞. To this we can associate its free (light) condensed derived abelian group $\mathbf{Z}[\mathbf{N} \cup \{\infty\}]$ and we will write $\mathbf{P}$ for the cofibre

$$\mathbf{Z}[\{\infty\}] \to \mathbf{Z}[\mathbf{N} \cup \{\infty\}] \to \mathbf{P}$$

of the canonical inclusion. Note that this is a split exact sequence. An important fact of this object is that it is internally compact in derived condensed abelian groups [CS24][Lecture 3].

2.1.4. One can use $\mathbf{P}$ to construct new affine analytic stacks out of old ones. Namely, if we consider an analytic ring A internally compactness of $\mathbf{P}$ formally implies that

$$\mathbf{P}_A := \mathbf{P} \otimes_{\mathbf{Z}} A \in \mathcal{D}(A)$$

is internally compact as well. Furthermore if we have given any collection W of endomorphisms of $\mathbf{P}_A$ and write

$$\widehat{(_)} \colon \mathcal{D}(A) \to \mathcal{D}(A)[W^{-1}]$$

for the Bousfield localization obtained by inverting maps in W. Then it is easy to check that

$$(A, \mathcal{D}(A)[W^{-1}])$$

represents a new affine analytic stack. All affine analytic stacks in this text will arise in this way.

2.1.5. The main example we will consider is the following. Consider an animated ring A together with some finitely generated ideal I in A and a subring A^+ of A. Then we will write

$$\mathrm{Spa}(A_{\widehat{I}}, A^+)$$

for the following construction. The (derived) I-adic completion of A as a condensed ring defines a solid ring, and we can invert the maps

$$f_\square := \mathrm{id} - f \cdot \mathrm{shift} \colon \mathbf{P}_{A_{\widehat{I}}} \to \mathbf{P}_{A_{\widehat{I}}}$$

for all $f \in A^+$ in $\mathrm{Mod}_{A_{\widehat{I}}}(\mathrm{Solid}(\mathbf{Z}))$ to obtain $\mathcal{D}(\mathrm{Spa}(A_{\widehat{I}}, A^+))$.

Remark 2.1.6. The analytic stack $\mathrm{Spa}(A, A^+)$ does not change if one adjoints, in A, topological nilpotent or elements to A^+. It also does not change after adjointing elements which are integral over A^+.

Remark 2.1.7. In the very end of this text, we will also consider a version of 2.1.5, but inverting topological nilpotent elements in A (i.e. consider derived Tate-Huber pairs).

Remark 2.1.8. In the case $A = A^+$, we will just write $\mathrm{Spa}(A)$.

2.1.9. Given a triple as in 2.1.5, we will write

$$|\mathrm{Spa}(A, A^+)|$$

for the space of continuous valuations on $\pi_0 A$, which value all functions in A^+ as ≤ 1. This can be understood as a quasi-compact topological space [Wed19][4.7]. And by [CS24][Lecture 9] there is a map of locales

$$\mathrm{Sm}(\mathcal{D}(\mathrm{Spa}(A, A^+))) \to |\mathrm{Spa}(A, A^+)|$$

where on the left we consider the smashing spectrum of a stable symmetric monoidal category [CS22][5.3].

2.1.10. By [CS24][Lecture 16], there is a six-functor formalism on the category of affine analytic stacks. One can then use this six-functor formalism to implement some aspects of geometry into affine analytic rings.

- A map of $j\colon U \to X$ of affine analytic stacks is called an open immersion if

$$j^*\colon \mathcal{D}(X) \to \mathcal{D}(U)$$

  admits a left adjoint $j_!$ and $j_!\mathcal{O}_U$ defines an idempotent coalgebra.
- A map $f\colon X \to S$ of affine analytic stacks is called proper if

$$\mathcal{D}(X) \simeq \mathrm{Mod}_{f_*\mathcal{O}_X}(\mathcal{D}(S)).$$

- A map of affine analytic stacks is called !-able, if it factors as an open immersion followed by a proper map.

Using this six-functor formalism, one can define a (Grothendieck) topology on affine analytic stacks. This topology is such that for a !-able covering $U \to X$ the functor $\mathcal{D}(_)^!$ (which implies the analogous statement for $\mathcal{D}(_)^*$) descends along this covering. Using the techniques of [HM24] one then obtains a six-functor formalism on the category of analytic stacks. We will write

$$\mathrm{AnStack}$$

for the latter.

Instead of going into the details of this defining topology, let us give the main examples, which will appear in this text.

Example 2.1.11. A cohomologically étale morphism of affine analytic stacks defines a covering if $\mathcal{D}(_)^*$ descends along this morphism.

Example 2.1.12. A proper morphism $f\colon X \to S$ of affine analytic stacks gives a covering if the algebra

$$f_*\mathcal{O}_X \in \mathcal{D}(S)$$

is descendable in the sense of [Mat16][3.18].

Example 2.1.13. Consider a map of affine analytic stacks $f\colon \mathrm{Spa}(B, B^+) \to \mathrm{Spa}(A, A^+)$. Then f is proper if and only if the map $A^+ \to B^+$ is integral up to topological nilpotent elements.

2.2. **Formal stacks and the éid-topology.** In this section, we will clarify what we mean by a formal stack and introduce the main topology used in this text.

Construction 2.2.1. Consider an animated ring A and a finitely generated ideal $I \subset \pi_0 A$, such that A is derived I-adically complete. We define a presheaf on the category of affine derived schemes $\mathrm{Spf}(A)$ by the cartesian square

$$\begin{array}{ccc} \mathrm{Spf}(A) & \longrightarrow & \mathrm{Spec}(A) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{\mathrm{cont}}(\pi_0 A, _) & \longrightarrow & \mathrm{Spec}(\pi_0 A) \end{array}$$

where we equip $\pi_0 A$ with the I-adic topology.

Definition 2.2.2. We will call a presheaf of the form $\mathrm{Spf}(A)$ an (derived) affine formal scheme.

Remark 2.2.3. Note that $\mathrm{Spf}(A)$ just depends on the I-adic topology on $\pi_0 A$ not on the ideal itself. Any ideal defining the same topology will be called an *ideal of definition*.

Remark 2.2.4. The category $\mathcal{D}(\mathrm{Spf}(A))$ of quasi-coherent sheaves on $\mathrm{Spf}(A)$ is computed by the full subcategory

$$\mathcal{D}_{I\text{-comp}}(A) \subset \mathcal{D}(A)$$

of those A-modules, which are derived I-adically complete.

2.2.5. Many types of maps between formal stacks can be defined by declaring them to be representable by schemes and then referring to the corresponding notion for schemes. We will use the following notions in this sense:

(a) *Affine open immersions.*
(b) *Affine étale morphisms.*
(c) *Integral morphisms.*

To define the topology we want to work with, we will make use of the notion of a descendable algebra from [Mat16].

Definition 2.2.6. A map of affine formal schemes $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$, which is representable by affine schemes, is called *descendable*, if

$$B \in \mathcal{D}(\mathrm{Spf}(A))$$

is a descendable algebra.

Construction 2.2.7. Consider a presheaf S on affine derived schemes. Then let us write

$$\mathrm{fSch}^{\mathrm{aff}}_{/S}$$

for the full subcategory of those presheaves over S, which are given by an affine formal scheme

$$\mathrm{Spf}(A) \to S,$$

such that the structure map as well as the diagonal $\mathrm{Spf}(A) \to \mathrm{Spf}(A) \times_S \mathrm{Spf}(A)$ are representable by affine schemes.

We will equip this category with a (pre)-topology called the *éid*-topology. This topology is the union of the étale topology with the topology generated by integral descendable maps. The induced topos will be denoted by

$$S_{\text{éid}}.$$

Variant 2.2.8. For later use, let us also record the following variant of 2.2.7. Assume S is a static p-nilpotent ring. Then we consider the category of static S-algebras and equip this category with the intersection of the flat topology with the éid-topology. We will write

$$S_{\text{éfid}}$$

for the induced topos.

Remark 2.2.9. Let us say a map $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$ is called *adic*, if there exists an ideal of definition I_A for $\mathrm{Spf}(A)$, such that

$$I_A \pi_0 B \subset \pi_0 B$$

gives an ideal of definition for $\mathrm{Spf}(B)$. Then a map of affine formal schemes is adic if and only if it is representable by affine schemes.

Remark 2.2.10. The ultimate reason we fix a base is the following: Consider a cospan of representable morphisms

$$\mathrm{Spf}(B) \to \mathrm{Spf}(A) \leftarrow \mathrm{Spf}(C).$$

Then the solid tensor product

$$C_{\widehat{I}} \otimes^{\square}_{A_{\widehat{I}}} B_{\widehat{I}}$$

is derived I-adically complete [Bos23][A.3]. This is not true if we do not assume the maps to be representable.

The generality of our base is chosen, as the objects we want to consider as a base will be honest stacks.

Remark 2.2.11. Any (derived) formal scheme X admitting a map to S, which locally on X in the Zariski topology is representable by affines, defines an object in $S_{\text{éid}}$. We will say a map of formal stacks over S is *étale* if it is representable by étale maps of formal schemes.

Remark 2.2.12. Note that associating to an affine formal stack $\mathrm{Spf}(A)$ its category of quasi-coherent sheaves $\mathcal{D}(\mathrm{Spf}(A))$ defines a sheaf on $S_{\text{éid}}$. In particular, we obtain a category of quasi-coherent sheaves $\mathcal{D}(X)$ for any formal stack X over S.

We will also use the following notion of covering.

Definition 2.2.13. We will say a map $f\colon X \to S$ of formal stacks is a descendable cover if

$$f_* \mathcal{O}_X \in \mathcal{D}(S)$$

is descendable.

We will use the following notions of maps of formal stacks and formal schemes.

Definition 2.2.14. We will say a map of affine formal schemes $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$ is of $^+$*finite type*, if it can be factored

$$\mathrm{Spf}(B) \to \mathbf{A}^n_{\mathrm{Spf}(A)} \to \mathrm{Spf}(A)$$

as an integral map followed by a projection from an affine space.

We will say a map $X \to S$ of p-adic formal schemes is:

- *$^+$Proper*, if the following conditions hold.
  (a) Zariski locally on target and domain it is of $^+$finite type.
  (b) It is represented by separated and universally closed morphisms of schemes.
- *Proper*, if it is $^+$proper and locally of finite presentation.

Furthermore, we will say a map of formal stacks $X \to S$ is *locally of $^+$finite type*, if, locally in the éid-topology on source and target, it is of $^+$finite type.

2.3. **From formal stacks to analytic stacks.** We will construct a functor from a suitable category of formal stacks to analytic stacks, and explain why this functor behaves well with respect to the geometries on both sides.

Remark 2.3.1. In the following, we will give criteria for proving that maps of formal stacks induce !-able or proper maps on the side of analytic stacks. For this, it will be convenient to work over an affine base. Note that there is no harm in doing so, as we can check !-ability as well as properness locally on the target.

2.3.2. To an affine formal scheme $\mathrm{Spf}(A)$, we associate the analytic stacks

$$\mathrm{Spa}(A) \to \mathrm{Spa}(A, \widetilde{\mathbf{Z}})$$

together with the canonical map coming from solidifying the points in A. This construction is natural in $\mathrm{Spf}(A)$.

Theorem 2.3.3. *The following hold:*

A *There exist left exact and colimit preserving functors*

$$(_)^{\square} \to (_)^{\square_{lp}} \colon S_{\acute{e}id} \to AnStack_{/S^{\square}}$$

together with a natural transformation. This datum is uniquely determined by the assertion that on affine formal schemes it is given by 2.3.2 and then one pulls back the target along the map $S^{\square} \to S^{\square_{lp}}$.

B *The functor $(_)^{\square}$ preserves étale morphisms and sends maps represented by integral maps of affine formal schemes to weakly proper morphisms.*

C *Given a map $X \to S = Spf(A)$ of formal stacks for which we can find a descendable cover*

$$Spf(B) \to X$$

from an affine formal scheme and which is locally on the source and target of $^+$finite type, the induced map

$$X^{\square} \to S^{\square}$$

is !-able.

D *Under the assumptions of **C** then the following are equivalent:*
  (a) The map $X^{\square} \to S^{\square}$ is co-smooth.
  (b) The map $X \times_S Spec(A/I) \to Spec(A/I)$ is co-smooth, where A/I is the quotient of A by some finitely generated ideal of definition[7].

[7]Here, this quotient depends on a chosen collection of finitely many generators of I. The conclusion of the statement, though, is independent of this choice.

E *Under the assumptions of* ***C*** *the map*

$$X^{\square} \to S^{\square}$$

is co-smooth, if and only if it is co-smooth over a finite Zariski stratification of S.

Before proving the theorem let us record the following corollary.

Corollary 2.3.4. *Consider a $^+$proper map $X \to S$ of p-adic formal schemes. Then the induced map*

$$X^{\square} \to S^{\square}$$

is weakly proper.

Proof. Note that the diagonal of a $^+$proper morphism is a closed immersion and thus weakly proper by 1.1.13 and 2.1.13. Thus by 1.1.12 we just have to see that the map is co-smooth. To see this, we note that such a map is quasi-compact, and we can find a descendable cover using a Zariski cover. In particular using 2.3.3[**D**] it suffices to check the claim after base change to $\mathrm{Spec}(\mathbf{F}_p)$. This base change now is a map of discrete adic spaces, so the claim follows from [Man22][2.9.29]. □

The proof of this Theorem will occupy the rest of the section. Let us start with the first paragraph.

The left exactness claims follow from 2.2.10 together with the observation that a map between affine formal schemes in $S_{\text{éid}}$ is representable by affine schemes (i.e. adic). Thus, to prove the first claim, we have to argue that the functors $(_)^{\square}$ and $(_)^{\square,\mathrm{lp}}$ send éid-coverings to effective epimorphisms in analytic stacks.

2.3.5. We will use the following observation in two places below. Any étale morphism $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$ of affine formal stacks sits in a cartesian square

$$\begin{array}{ccc} \mathrm{Spf}(B) & \longrightarrow & \mathrm{Spf}(A) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(B') & \longrightarrow & \mathrm{Spec}(A) \end{array}$$

where the lower horizontal map is an étale morphism of schemes. Furthermore if the map of formal schemes was surjective, the map of schemes can be chosen to be surjective as well.

This follows from [Sta, Tag 0AN8] combined with the fact that, for an animated ring A, base change along the map $A \to \pi_0 A$ induces an equivalence on étale algebras.

Proof of 2.3.3 ***A*** $(_)^{\square,\mathrm{lp}}$. Consider an étale morphism $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$ of formal schemes. Using 2.3.5 together with the fact that faithfully flat étale maps of animated rings are descendable [Mat16][3.33], we see that

$$B \in \mathcal{D}(\mathrm{Spf}(A))$$

is a descendable algebra. Now, as $(_)^{\square,\mathrm{lp}}$ sends maps of affine formal schemes to proper analytic ring maps 2.1.13, we have to see that an éid-covering gets sent to a descendable map of algebras 2.1.12. But there is a lax monoidal functor

$$\mathcal{D}(\mathrm{Spf}(A)) \to \mathcal{D}_{I\text{-comp}}(\mathcal{D}(\mathrm{Spa}(A, \widetilde{\mathbf{Z}}))) \subset \mathcal{D}(\mathrm{Spa}(A, \widetilde{\mathbf{Z}}))$$

and such functors preserve descendable algebras [BS17][11.20]. □

The following will also prove the second paragraph of the proposition.

*Proof of 2.3.3 **A** and **B** $(_)^{\square}$.* We first do the case of an integral descendable cover. Using the same argument as in the proof for $(_)^{\square_{\mathrm{lp}}}$, it is enough to check that $(_)^{\square}$ sends integral maps to proper maps of analytic rings. As for any solid affinoid A, the topological nilpotent elements are contained in A^+ [CS24][Lecture 8], it suffices to check that

$$\widetilde{A + IB} = B$$

where I is some ideal of definition for A. But given a point $b \in B$, by assumption, we can find a monic polynomial f over A such that

$$f(b) = i \in IB.$$

This means that

$$b \in \widetilde{A[i] + IB} = \widetilde{A + IB}$$

and we win.

For the case of an étale morphism, consider a cartesian square as in 2.3.5 and write $\mathrm{Spa}(A)^{\mathrm{disc}}$ for the discrete adic space associated to the underlying discrete ring of A (and similar for B). Then we have a commutative square

$$\begin{array}{ccc} \mathrm{Spa}(B) & \longrightarrow & \mathrm{Spa}(A) \\ \downarrow & & \downarrow \\ \mathrm{Spa}(B')^{\mathrm{disc}} & \xrightarrow[j]{} & \mathrm{Spa}(A)^{\mathrm{disc}}. \end{array}$$

Now it is proven in [CS19][11.4, 11.6] that the lower horizontal map is étale. In particular, the functor j^* commutes with limits, which implies that the square pictured above is cartesian. This implies that the upper horizontal map is étale as well, and to see that it is an epimorphism in analytic stacks (in the case the map of formal schemes was surjective), we have to check that the functor $\mathcal{D}^*$ descends along the map j. This is shown in [Man22][2.10.6]. □

*Proof of 2.3.3 **C**.* We first claim that the map

$$(X/S)^{\square_{\mathrm{lp}}} \to S^{\square}$$

is co-smooth.

Using the descendable cover, we obtain a proper descendable cover

$$(\mathrm{Spf}(B)/S)^{\square_{\mathrm{lp}}} \to (X/S)^{\square_{\mathrm{lp}}}$$

of analytic stacks. By 1.1.11[(c)] it now suffices to see that the composition

$$(\mathrm{Spf}(B)/S)^{\square_{\mathrm{lp}}} \to S^{\square}$$

is co-smooth. This is even weakly cohomologically proper 2.1.13.

Under the $^+$finite type assumption, we claim that the map

$$X^{\square} \to (X/S)^{\square_{\mathrm{lp}}}$$

is an open immersion and thus !-able. This implies the !-ability claim. To see this, we can work locally on the target and thus assume

$$X \simeq \mathrm{Spf}(B) \to \mathbf{A}^n_S \to S$$

is affine and factors as an integral map followed by the projection from some affine space. In this case, the map in question is obtained by solidifying the coordinates and thus an open immersion. □

To prove assertion **D**, we will need to associate a topological space to a formal stack. For this recall from 2.1.9 that, to an affine formal scheme $\mathrm{Spf}(B)$ over $\mathrm{Spf}(A)$, we can associate two topological spaces

$$|\mathrm{Spa}(B)| \text{ and } |\mathrm{Spa}(B, \widetilde{A})|.$$

We will need the following lemma.

Lemma 2.3.6. For any éid-cover $\mathrm{Spf}(C) \to \mathrm{Spf}(B)$ of affine formal schemes over an affine formal scheme $\mathrm{Spf}(A)$, the induced maps

$$|\mathrm{Spa}(C)| \to |\mathrm{Spa}(B)| \text{ and } |\mathrm{Spa}(C, \widetilde{A})| \to |\mathrm{Spa}(B, \widetilde{A})|$$

are submersive.

Proof. We have already seen in **A** and **B** that the upper-$*$-functors on these analytic stacks are conservative. This implies that the maps in question are surjective. Now such an éid-cover is given by a finite composition of either étale covers or integral descendable covers. Thus by [Sta, Tag 02YB] and [Sta, Tag 0AAU] it suffices to show that each of those is either open (in the case of an étale cover) or closed (in the case of an integral map). The case of an étale map is [Hub96][1.7.9], but we will give a similar argument in the case of an integral map. For this case, we will explain the map

$$|\mathrm{Spa}(C)| \to |\mathrm{Spa}(B)|$$

as the argument for the other map is the same.

Note first that up to completion, we can write C as a filtered colimit of finite B-algebras. Thus, as the adic spectrum just depends on the completion, we can assume that $B \to C$ is integral. Now as in [KH94], we write $\mathrm{Spv}(B)$ for the spectrum of all valuations on B and similar for C. Then by [KH94][2.1.7.(ii)] the map $\mathrm{Spv}(C) \to \mathrm{Spv}(B)$ is closed. Furthermore, let us write

$$\mathrm{Spv}(B)^+ \text{ and } \mathrm{Spv}(B, B^{\circ\circ})^+$$

for the subspace of those valuations which value any function of B as ≤ 1 and the subspace of those valuations, which, in addition, value any topological nilpotent element as < 1. Then we have a Cartesian square

$$\begin{array}{ccc} \mathrm{Spv}(C)^+ & \longrightarrow & \mathrm{Spv}(C) \\ \downarrow & & \downarrow \\ \mathrm{Spv}(B)^+ & \longrightarrow & \mathrm{Spv}(B) \end{array}$$

which shows that the left vertical map is also closed. Furthermore the subspace

$$\mathrm{Spv}(B, B^{\circ\circ})^+ \subset \mathrm{Spv}(B)^+$$

is closed (and similar for C), which shows that also the map $\mathrm{Spv}(C, C^{\circ\circ})^+ \to \mathrm{Spv}(B, B^{\circ\circ})^+$ is closed. By [Hub93][2.6] there exist retractions making the following

diagram commute

$$\begin{array}{ccccc} \mathrm{Spa}(C) & \xrightarrow{i_C} & \mathrm{Spv}(C, C^{\circ\circ})^+ & \xrightarrow{r_C} & \mathrm{Spa}(C) \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} \\ \mathrm{Spa}(B) & \xrightarrow[i_B]{} & \mathrm{Spv}(B, B^{\circ\circ})^+ & \xrightarrow[r_B]{} & \mathrm{Spa}(B). \end{array}$$

From this we see that for any closed $Z \subset \mathrm{Spa}(C)$, we can write

$$f(Z) = g(r_C^{-1}(Z)) \cap \mathrm{Spa}(B)$$

which is closed. □

Construction 2.3.7. Using 2.3.6, we see that associating to an affine formal scheme $\mathrm{Spf}(B)$ over $\mathrm{Spf}(A)$ the topological spaces $|\mathrm{Spa}(B)|$ and $|\mathrm{Spa}(B, \widetilde{A})|$ induces colimit preserving functors

$$|(_)^{\square}| \text{ and } |(_)^{\square_{\mathrm{lp}}}| \colon A_{\text{éid}} \to \mathrm{Top}$$

.

2.3.8. Recall from 2.1.9 that for any affine formal scheme $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$, we had maps of locales

$$\mathrm{SmSpec}(\mathcal{D}(\mathrm{Spa}(B))) \to |\mathrm{Spa}(B)| \text{ and } \mathrm{SmSpec}(\mathcal{D}(\mathrm{Spa}(B, \widetilde{A}))) \to |\mathrm{Spa}(B, \widetilde{A})|.$$

Using those maps, we obtain maps of locales

$$\mathrm{SmSpec}(\mathcal{D}(X^{\square})) \to |X^{\square}| \text{ and } \mathrm{SmSpec}(\mathcal{D}(X^{\square,\mathrm{lp}})) \to |X^{\square_{\mathrm{lp}}}|$$

for any formal stack X over $\mathrm{Spf}(A)$.

Proof of 2.3.3 ***D****.* Arguing as in the proof of **C**, we can factor the map as

$$X^{\square} \to (X/S)^{\square} \to S^{\square}$$

an open immersion followed by a proper map. So it suffices to see that the open immersion is an isomorphism. This open immersion comes from an open immersion

$$|X^{\square}| \subset |(X/S)^{\square_{\mathrm{lp}}}|$$

and we first claim that it suffices to show that the latter map is surjective. Indeed, the open immersion $X^{\square} \to (X/S)^{\square_{\mathrm{lp}}}$ corresponds to an idempotent algebra in $\mathcal{D}((X/S)^{\square_{\mathrm{lp}}})$ and is an isomorphism if and only if this idempotent algebra is 0. But by the surjectivity assumption and 2.3.8, we can check this after pulling back along the map itself, where the idempotent algebra vanishes by construction. To see that this inclusion is surjective, we choose a descendable cover

$$\mathrm{Spf}(B) \to X$$

and pull the open subset back along the map

$$\mathrm{Spa}(B, \widetilde{A}) \to (X/S)^{\square_{\mathrm{lp}}}.$$

This way we obtain an open subset $U \subset \mathrm{Spa}(B, \widetilde{A})$ which induces an isomorphism on non-Tate points. The next lemma shows that any point in the target specializes to such a point, which implies that it is an isomorphism itself. □

We used the following lemma.

Lemma 2.3.9. Consider a ring A which is I-adically complete for some finitely generated ideal $I \subset A$. Furthermore, let $A^+ \subset A$ be an integrally closed subring that contains the generators of I. Then any point in $\mathrm{Spa}(A, A^+)$ specializes to a non-Tate point.

Proof. Let us be given a point $x \in \mathrm{Spa}(A, A^+)$ corresponding to a valuation

$$x \colon A \to \Gamma_x \cup \{0\}.$$

Then we write $c\Gamma_x \subset \Gamma_x$ for the convex closure of $\mathrm{Im}(x) \cap \Gamma_{x,\geq 1}$ and consider the map

$$x_{|c\Gamma_x} \colon A \to c\Gamma_x \cup \{0\}, |f(x)| = \begin{cases} |f(x)| & \text{if } |f(x)| \in c\Gamma_x \\ 0 & \text{if } |f(x)| \notin c\Gamma_x. \end{cases} \tag{2.3.1}$$

As the subgroup is convex, this defines a valuation [Wed19][4.15], and it clearly values any function $f \in A^+$ with ≤ 1. As it sends any topological nilpotent element to 0, it is continuous and defines a non-Tate point. It is a specialization of x by [Wed19][4.16]. □

Proof of 2.3.3 ***E****.* Using **D**, we can assume that the map

$$X^\square \to S^\square$$

is a map of discrete adic spaces. For an affine discrete adic space $\mathrm{Spa}(A, A^+)$, taking the support

$$|\mathrm{Spa}(A, A^+)| \to |\mathrm{Spec}(A)|$$

is compatible with Zariski-closed and Zariski-open sets. So we can argue similarly to the proof of **D**. □

3. The solid syntomification

3.1. **The infinite roots topology.** We will introduce a topology on the category of p-adic formal schemes called the *infinite p-root topology*. This topology will be weaker than the quasi-syntomic topology originally used in the theory of prismatic and syntomic cohomology [BMS18] [BL22a], but still strong enough to make most of the local to global arguments work. On the other hand, it will induce surjections of prismatic stacks on the level of analytic stacks.

By an affine p-adic formal scheme, we mean an affine (derived) formal scheme, which is adic over $\mathrm{Spf}(\mathbf{Z}_p)$. The main example to have in mind is the following.

Example 3.1.1. For any set I, we will call a map of the form

$$\mathrm{Spf}(\mathbf{Z}_p[x_i^{\frac{1}{p^\infty}} | i \in I] \to \mathrm{Spf}(\mathbf{Z}_p[x_i | i \in I])$$

a generic infinite p-root cover.

Definition 3.1.2. A family of maps $\{\mathrm{Spf}(B_i) \to \mathrm{Spf}(A)\}$ of p-adic formal schemes is called an *infinite p-root cover*, if it can be written as a finite composition of one of the following:

(1) $\mathrm{Spf}(A)$ is the finite disjoint union of the $\mathrm{Spf}(B_i)$.
(2) The family is given by one map $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$, which is pulled back from a generic infinite p-root cover 3.1.1.

Lemma 3.1.3. The collection of infinite p-root covers forms a Grothendieck pretopology on the category of affine p-adic formal schemes.

Proof. We have to see that the collection of those covers contains all identities and is stable under composition and pullback. The first assertion is clear, and the others hold by definition. □

We will write $\mathrm{Sh}_{\sqrt{\infty}}(\mathrm{fSch}^{\mathrm{aff}}_{\mathbf{Z}_p})$ for the induced topos.

Remark 3.1.4. A presheaf on p-adic formal schemes is a sheaf for the infinite p-root topology, if and only if it sends finite disjoint unions to products and satisfies the Čech condition for maps pulled back from generic infinite p-root covers.

In the case we want to construct a map of topoi

$$\mathrm{Sh}_{\sqrt{\infty}}(\mathrm{fSch}^{\mathrm{aff}}_{\mathbf{Z}_p}) \to \mathcal{X}$$

starting from a finite limit-preserving functor from (affine) p-adic formal schemes to $\mathcal{X}$. The latter condition will simplify. Namely, we only have to check the Čech condition on generic infinite p-root covers.

The following easy observation will be crucial for us.

Lemma 3.1.5. Consider a map $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$, which is a finite composition of maps pulled back from generic infinite p-root covers. Then we have the following:

(1) The map $A/p \to B/p$ is integral.
(2) The cofibre of the map $A/p \to B/p$ is a projective A/p-module.
(3) The shifted cotangent complex $\mathcal{L}_{B/A}/p[-1]$ is projective.

Proof. All assertions are stable under finite compositions and base change. So we can check them for a generic infinite p-root cover. Then (1) and (2) are clear and (3) follows, using the fundamental sequence of the cotangent complex, from the observations that $A/p \to B/p$ is the colimit perfection of the domain and that the cotangent complex of perfect $\mathbf{F}_p$-algebras vanishes. □

We can also describe a basis for this topology.

Definition 3.1.6. A p-complete animated ring S is called *semiperfectoid*, if there exists a π_0-surjection $R \to S$ from a perfectoid ring R. We will write

$$\mathrm{SemPerfd}^{\mathrm{aff}}_{\mathbf{Z}_p} \subset \mathrm{fSch}^{\mathrm{aff}}_{\mathbf{Z}_p}$$

for the full subcategory of those affine p-adic formal schemes, whose functions form a semiperfectoid ring.

Remark 3.1.7. One natural way to define animated perfectoid rings is as quotients A/I where $I \to A$ is a perfect animated prism. Note that all such are static [BL22c][2.16].

Lemma 3.1.8. The category of affine semiperfectoid p-adic formal schemes forms a basis of the infinite p-root topology on affine p-adic formal schemes. In particular, restricting along the inclusion induces an equivalence of topoi

$$\mathrm{Sh}_{\sqrt{\infty}}(\mathrm{fSch}^{\mathrm{aff}}_{\mathbf{Z}_p}) \simeq \mathrm{Sh}_{\sqrt{\infty}}(\mathrm{SemPerfd}^{\mathrm{aff}}_{\mathbf{Z}_p}).$$

Proof. This follows the same way as in [BMS18][4.28] using 3.1.1. □

3.1.9. We will write $\sqrt{\infty}^{\text{ét}}$-topology for the union of the infinite p-root topology with the étale topology. We will consider this topology on the category of qcqs (derived) p-adic formal schemes. That is, those formal schemes that are adic over $\mathrm{Spf}(\mathbf{Z}_p)$. The corresponding topos will be denoted by

$$\mathrm{Spf}(\mathbf{Z}_p)_{\sqrt{\infty}^{\text{ét}}}.$$

Clearly, this topos is also generated by semiperfectoid p-adic formal schemes.

3.2. **W-modules.** We will collect some examples of W-modules and recall some facts about them.

3.2.1. Note that associating to an affine p-adic formal scheme its Witt-vectors defines a sheaf of (p-complete) animated rings on the éid-topology. To see this, we can write

$$W \simeq \lim_n W_n$$

as the limit of the truncated Witt-vectors and then recall that we have an isomorphism

$$W_n \simeq \mathbf{A}^n$$

as presheaves of anima. So the claim follows as the éid-topology is subcanonical.

3.2.2 (**$\mathbf{G}_{\mathrm{a}}^{\#}$-bundles**)**.** For an animated ring R and a connective R-module $\mathcal{E}$, we will write

$$\Gamma_R(\mathcal{E})$$

for the *animated divided power algebra* over R associated to $\mathcal{E}$[8]. We then will write

$$\mathbf{V}(\mathcal{E})^{\#}_R := \mathrm{Spec}(\Gamma_R(\mathcal{E})).$$

Example 3.2.3. In the case the $\mathbf{G}_{\mathrm{a}}^{\#}$-bundle is associated to the trivial vector bundle of rank one, we obtain the following. Let $\mathbf{G}_{\mathrm{a}}^{\#}$ be the PD-hull of the origin in $\mathbf{G}_{\mathrm{a}}$ over $\mathbf{Z}$. Concretely we have

$$\mathbf{G}_{\mathrm{a}}^{\#} \simeq \mathrm{Spec}(\mathbf{Z}[x, \frac{x^2}{2!}, \frac{x^3}{3!}, \dots]).$$

Note that there is a map $\mathbf{G}_{\mathrm{a}}^{\#} \to \mathbf{G}_{\mathrm{a}}$ and that the equalities

- $\frac{(x+y)^n}{n!} = \sum_{i+j=n} \frac{x^i}{i!}\frac{y^j}{j!}$
- $\frac{(xy)^n}{n!} = \frac{x^n}{n!} \cdot y^n$

show that the additive group action on $\mathbf{G}_{\mathrm{a}}$ induces an action on $\mathbf{G}_{\mathrm{a}}^{\#}$ and the the multiplicative group action on $\mathbf{G}_{\mathrm{a}}$ induces via the above map an action on $\mathbf{G}_{\mathrm{a}}^{\#}$ which makes it into a $\mathbf{G}_{\mathrm{a}}$-module scheme. In particular we can understand $\mathbf{G}_{\mathrm{a}}^{\#}$ as a W-module scheme via the projection $W \to \mathbf{G}_{\mathrm{a}}$.

The following in particular shows that W stays a sheaf if we let it take values in possibly non-connective spectra.

[8]This can be obtained by animating the classical construction from pairs $(R, \mathcal{E})$ where R is a polynomial algebra in finitely many variables and $\mathcal{E}$ a finite free R-module. The animation of the category of such pairs gives the total category of the fibration corresponding to the functor which assigns to an animated ring its category of connective modules.

Lemma 3.2.4. Given an affine p-adic formal scheme $\mathrm{Spf}(S)$ and a finite projective $W(S)$-module P, we have

$$R\Gamma_{\text{éid}}(S, P \otimes_{W(S)} W) \in \mathcal{D}(W(S))_{\geq 0}$$

and the same holds for the truncated Witt vectors.

Proof. Writing P as a summand of a finite free $W(S)$-module this reduces to showing $R\Gamma(S, W) \in \mathcal{D}(W(S))_{\geq 0}$. Furthermore writing W as a limit of the truncated Witt Vectors and observing that the transition maps are surjective, the Milnor sequence tells us that this reduces to the claim

$$R\Gamma(S, W_n) \in \mathcal{D}(W(S))_{\geq 0}.$$

This follows by induction on n using the fiber sequences $W_{n-1} \to W_n \to \mathbf{G}_{\mathrm{a}}$ coming from the Verschiebung and the corresponding claim for $\mathbf{G}_{\mathrm{a}}$. □

For an affine p-adic formal scheme $\mathrm{Spf}(S)$, we will write

$$\mathcal{D}_{\text{éid}}(\mathrm{Spf}(S), W)$$

for the category of W-modules in sheaves of spectra on the éid-topos of $\mathrm{Spf}(S)$.

Lemma 3.2.5. [Dri24][3.4] [BL22b][2.6.1] [BL22a][3.4.11] The Frobenius $F\colon W \to F^*W$ is a π_0 surjection in the éid-topology. Furthermore, if we write $W[F]$ for the fibre of the Frobenius, the composition $W[F] \subset W \to \mathbf{G}_{\mathrm{a}}$ of the inclusion with the projection lifts uniquely to an isomorphism

$$W[F] \simeq \mathbf{G}_{\mathrm{a}}^{\#}.$$

In particular, there is a fiber sequence

$$\mathbf{G}_{\mathrm{a}}^{\#} \to W \to F_*W$$

in $\mathcal{D}_{\text{éid}}(\mathrm{Spf}(S), W)$ where the second map is given by the Frobenius.

Proof. The proof of [BL22b][2.6.1] shows that the Frobenius is faithfully flat; thus it is descendable by [Mat16][3.31] as the domain is countable. It is also clearly integral. This shows π_0-surjectivity.

For the rest, note that the ordinary scheme represents $W[F]$ also on derived schemes

$$\mathrm{Spec}(\mathbf{Z}_{(p)}[x_0, x_1, x_2, \dots]/(x_0^p + px_1, x_1^p + px_2, \dots))$$

and the proof of [BL22b][2.6.1] gives an isomorphism $W[F] \simeq \mathbf{G}_{\mathrm{a}}^{\#}$ of representing objects. □

Corollary 3.2.6. *[BL22c][5.7] For any affine p-adic formal scheme $Spf(S)$ and any finite projective $W(S)$-module P there are fibre sequences*

- $R\Gamma_{\acute{e}id}(Spf(S), P \otimes_W W_n[F]) \to P \otimes_W W_n(S) \to P \otimes_W F_*W_{n-1}(S)$ $(n \geq 2)$.
- $R\Gamma_{\acute{e}id}(Spf(S), P \otimes_W W[F]) \to P \otimes_W W(S) \to P \otimes_W F_*W(S)$

in $\mathcal{D}(W(S))$.

Proof. We claim that these fiber sequences arise as the global sections of the fiber sequence

$$W[F] \to W \to F_*W$$

tensored with P (and the analogous fiber sequence with the truncated Witt Vectors). That is the essential claim is that $R\Gamma(\mathrm{Spf}(S), P \otimes_{W(S)} W) \in \mathcal{D}(W(S))_{\geq 0}$. This was explained in 3.2.4. □

Corollary 3.2.7. *For any affine p-adic formal scheme, there is a commutative diagram*

$$\begin{array}{ccccc} & & F_*W & = & F_*W \\ & & \downarrow{\scriptstyle V} & & \downarrow{\scriptstyle p} \\ W[F] & \longrightarrow & W & \xrightarrow{F} & F_*W \\ \| & & \downarrow & & \downarrow \\ \mathbf{G}_{\mathrm{a}}^{\#} & \longrightarrow & \mathbf{G}_{\mathrm{a}} & \longrightarrow & \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}} \simeq F_*W/p \end{array}$$

in $\mathcal{D}_{éid}(Spf(S), W)$. Such that the square in the middle is cartesian and the horizontal and vertical sequences are (co)fiber sequences.

Proof. By 3.2.5, we only have to give the identification in the lower right corner. This is explained in [BL22b][2.6.8]. □

For the following we consider a static p-nilpotent ring S and recall the topos $S_{\text{éfid}}$ from 2.2.8.

Lemma 3.2.8. Let S be a static p-nilpotent ring. Then the flat éid-topology on S admits a basis of rings T, such that the maps

$$F\colon W_n(T) \to F_*W_{n-1}(T)$$

are surjective for all $n \geq 2$.

Proof. In [DK14][3.2] the authors explain that for this surjectivity to hold it is enough that the Frobenius $F\colon W(T) \to F_*W(T)$ on the whole Witt Vectors hits all multiplicative lifts $[t]$ of elements $t \in T$. As we have the formula $F([t]) = [t^p]$ this is true for any ring admitting all p-th roots of its elements.

Now consider a static p-nilpotent S-algebra S_0. Then we consider the countable chain

$$S_0 \to S_1 \to S_2 \to \cdots$$

where the maps $S_n \to S_{n+1}$ are obtained by freely adjoining p-th roots of all elements. Write $T = \mathrm{colim}_n S_n$; then the map $S_0 \to T$ is clearly flat and integral, and any element in T admits a p-th root. To see that it is descendable, note that for each n the fiber of the map $S_n \to S_{n+1}$ takes the form $\oplus_I S_n[-1]$ for some set I. From this one easily sees that the map $S_n \to S_{n+1}$ is descendable of index ≤ 0 [BS17][11.18], such that the map $S_0 \to T$ becomes descendable by [BS17][11.22]. □

3.3. **Recollections on prismatic stacks.** We will make use of the stacky approach to prismatic and syntomic cohomology invented by Bhatt-Lurie [BL22a] [BL22c] and Drinfeld [Dri24]. More references and details on this theory can be found in [BL22b] [GM24] [Hau25].

Let us recall some definitions. We will interpret the following objects as éid-sheaves (with values in anima) on affine p-adic formal schemes. Having said this, we will

describe the functor of points on p-nilpotent rings and then extend the presheaves by left Kan-extension. As an example, we obtain the formula

$$\mathbf{G}_{\mathrm{a}}(\mathrm{Spf}(A)) = A_{\widehat{p}}.$$

3.3.1 (**The de Rham stack**)**.** Recall the ring stack, which assigns to a p-nilpotent ring R the animated ring

$$\mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R) := \mathbf{G}_{\mathrm{a}}(R)/\,\mathbf{G}_{\mathrm{a}}^{\#}(R).$$

We then define the *de Rham stack* of a p-adic formal scheme X via transmutation along this ring stack. That is, the anima of R-valued points is given by

$$X^{dR}(R) := X(\mathrm{Spec}(\mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R))).$$

3.3.2 (**The Hodge stack**)**.** Recall the ring stack over $\mathrm{B}\mathbf{G}_{\mathrm{m}}$, which assigns to a p-nilpotent animated ring R with a line bundle $\mathcal{L}$ on $\mathrm{Spec}(R)$ the square zero extension

$$\mathbf{G}_{\mathrm{a}}^{\mathrm{Hodge}}(\mathrm{Spec}(R) \to \mathrm{B}\mathbf{G}_{\mathrm{m}}) := R \oplus B\mathbf{V}_R(\mathcal{L}^{-1})^{\sharp}(R).$$

For a p-adic formal scheme X, we then define the *Hodge stack* via transmutation. That is the anima of R-valued points for a p-nilpotent animated ring R is given by

$$X^{Hodge}(\mathrm{Spec}(R) \to \mathrm{B}\mathbf{G}_{\mathrm{m}}) := X(\mathrm{Spec}(\mathbf{G}_{\mathrm{a}}^{\mathrm{Hodge}}(R))).$$

3.3.3 (**The Prismatisation**)**.** For a p-nilpotent animated ring R, we will write

$$\mathbf{Z}_p^{\mathbb{\Delta}}(R)$$

for the anima of *Cartier-Witt divisors* on R as defined in [BL22c][8.2]. Then recall the ring stack over $\mathbf{Z}_p^{\mathbb{\Delta}}$, which assigns to a Cartier-Witt divisor $\alpha\colon I \to W(R)$ on R, the animated ring

$$\mathbf{G}_{\mathrm{a}}^{\mathbb{\Delta}}(\mathrm{Spec}(R) \to \mathbf{Z}_p^{\mathbb{\Delta}}) := W(R)/I.$$

To define the *Prismatisation* of a general p-adic formal scheme X, we again use transmutation. It assigns to a Cartier divisor on R the anima

$$X^{\mathbb{\Delta}}(\mathrm{Spec}(R) \to \mathbf{Z}_p^{\mathbb{\Delta}}) := X(\mathrm{Spec}(\mathbf{G}_{\mathrm{a}}^{\mathbb{\Delta}}(R))).$$

We already obtained the following.

Corollary 3.3.4. *For a Cartier-Witt divisor $I \to W(S)$, we have*

$$R\Gamma_{\acute{e}id}(S, \mathbf{G}_{\mathrm{a}}^{\mathbb{\Delta}}) \in \mathcal{D}(W(S))_{\geq 0}$$

and the latter admits the structure of an animated ring.

Proof. The first claim follows from 3.2.4 and the cofiber sequence

$$I \otimes_{W(S)} W \to W \to \overline{W}$$

of sheaves on $(S)_{\text{éid}}$ with values in $\mathcal{D}(W(S))$. For the second claim, note that the Cartier-Witt divisor can be seen as a map $\mathrm{Spec}(W(S)) \to \mathbf{A}^1/\,\mathbf{G}_{\mathrm{m}}$ and we can pull back this map to $\mathrm{B}\mathbf{G}_{\mathrm{m}}$. □

Example 3.3.5. Given an animated prism $I \to A$, there is a map

$$\mathrm{Spf}(A) \to \overline{A}^{\mathbb{\Delta}}$$

which is constructed as follows. A map $A \to S$, by adjunction, uniquely refines to a map $A \to W(S)$ of δ-rings. Then $I \otimes_A W(S) \to W(S)$ defines a Cartier-Witt divisor and we obtain a map $\overline{A} \to W(S)/I$ by pushing out along $A \to \overline{A}$. As explained in [BL22c][3.12], in the case the prism is perfect with corresponding perfectoid $\overline{A} \simeq R$ this construction gives an isomorphism of functors

$$\mathrm{Spf}(A) \simeq R^{\mathbb{\Delta}}.$$

Example 3.3.6. The moduli of Cartier-Witt divisors is essentially the moduli of animated prisms [BL22c][8.3]. An important example of a prism is the universal oriented prism. For an animated p-nilpotent ring R we write $W_0(R) \subset W(R)$ for the subspace of those Verschiebung expansions $\sum_{n\geq 0} V^n[a_n]$ for which a_0 is nilpotent and a_1 is a unit. Note that, as a functor, this assignment is represented by an affine formal scheme

$$\mathrm{Spf}(\mathbf{Z}_p\langle a_0, a_1^{\pm 1}, a_2, \dots\rangle)$$

where we also complete to (a_0), and that the Frobenius on the Witt Vectors restricts to this subspace, such that we obtain a δ-structure on the representing ring. Thus choosing (a_0) as an ideal we have produced a prism and as in [BL22a][3.2.4] we get a map

$$W_0 \to \mathbf{Z}_p^{\mathbb{\Delta}}.$$

By (the proof of) [BL22a][3.2.3] (see also [BL22c][8.5]), this map identifies the target as the quotient, in the Zariski topology, of the domain by the canonical action of the affine group scheme representing the functor $R \mapsto W(R)^\times$. Where we write $W(R)^\times$ for the units in the Witt Vectors. This group scheme is represented by the free delta ring on a unit $\mathbf{Z}_p\{u^{\pm}\}$.

In particular, we see that we have found a presentation of

$$\mathbf{Z}_p^{\mathbb{\Delta}} \in (\mathbf{Z}_p^{\mathbb{\Delta}})_{\text{éid}}.$$

Variant 3.3.7. As pointed out in 3.3.6, for any animated prism $I \to A$ there is a map

$$\mathrm{Spf}(A) \to \mathbf{Z}_p^{\mathbb{\Delta}}.$$

Pulling back the ring stack $\mathbf{G}_{\mathrm{a}}^{\mathbb{\Delta}}$ along this map gives a ring stack

$$(\mathbf{G}_{\mathrm{a}}/\mathrm{A})^{\mathbb{\Delta}} \to \mathrm{Spf}(A)$$

which via transmutation defines the relative Prismatisation over A for p-adic formal schemes over $\mathrm{Spf}(A/I)$.

3.3.8 (The Hodge-Tate stack). Base changing a Cartier-Witt divisor on a p-nilpotent ring R along the map $W(R) \to R$ produces a map

$$\mathbf{Z}_p^{\mathbb{\Delta}} \to \widehat{\mathbf{A}^1}/\,\mathbf{G}_{\mathrm{m}}\,.$$

The locus over $\mathrm{B}\mathbf{G}_{\mathrm{m}}$ is called the *Hodge-Tate* locus and will be denoted by

$$X^{HT}.$$

One can also define this stack via transmutation using the ring stack

$$\mathbf{G}_{\mathrm{a}}^{\mathrm{HT}} := \mathbf{G}_{\mathrm{a}}^{\mathbb{\Delta}}{}_{|\mathbf{Z}_p^{HT}}$$

over $\mathbf{Z}_p^{HT}$ which we will investigate in more detail later.

3.3.9 (**The Breuil-Kisin twist on tha Prismatisation**)**.** Let us write $\mathcal{O}(-1) \to \mathcal{O}$ for the universal generalized Cartier divisor on $\mathbf{A}^1/\mathbf{G}_{\mathrm{m}}$. Then pulling back the line bundle $\mathcal{O}(-1)$ via the map

$$\mathbf{Z}_p^{\mathbb{\Delta}} \to \widehat{\mathbf{A}^1}/\mathbf{G}_{\mathrm{m}}$$

defines a line bundle on $\mathbf{Z}_p^{\mathbb{\Delta}}$ which we will denote by $\mathcal{O}\{1\}$ an call the *Breuil-Kisin twist*. We will denote several pullbacks of this line bundle in the same way.

3.3.10 (**Filtered Cartier-Witt divisors**)**.** The moduli of Nygaard filtered Cartier-Witt divisors can be described as the following pullback (see [Hau25][2.3.9])

$$\begin{array}{ccc} \mathbf{Z}_p^{\mathcal{N}} & \xrightarrow{\quad\pi\quad} & \mathbf{Z}_p^{\mathbb{\Delta}} \\ {\scriptstyle (t,u)}\downarrow & & \downarrow{\scriptstyle \tilde{\mu}} \\ \mathbf{A}^1/\mathbf{G}_{\mathrm{m}} \times (\mathbf{A}^1/\mathbf{G}_{\mathrm{m}})^{dR} & \xrightarrow[m]{} & (\mathbf{A}^1/\mathbf{G}_{\mathrm{m}})^{dR} \end{array}$$

where the maps can be described as follows.

- The map m takes a pair of generalized Cartier divisors $(t\colon \mathcal{L}_1 \to R, u\colon \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R) \to \mathcal{L}_2)$ to the generalized Cartier divisor

  $$\mathcal{L}_1 \otimes_R \mathcal{L}_2 \to \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R)$$

  obtained by base changing t along the map $R \to \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R)$ and then precomposing with u^{-1}.
- To describe the map $\tilde{\mu}$, one starts with a Cartier-Witt divisor $I \to W(R)$ and considers its frobenius pullback $F_*I \to F_*W(R)$. Base changing the latter along the map

  $$F_*W(R) \to F_*W(R)/p \simeq \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R)$$

  (see [BL22b][2.6.8] for the used isomorphism) gives a generalized Cartier divisor

  $$F_*I \otimes_{F_*W(R)} \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R) \to \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R).$$

We will refer to π as the *structure map* and to t as the *Rees map*.

Thus for a p-nilpotent animated ring R, a Nygaard filtered Cartier-Witt divisor $\mathrm{Spec}(R) \to \mathbf{Z}_p^{\mathcal{N}}$ is given by a triple

$$(\mathcal{L}_1 \to R, \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R) \to \mathcal{L}_2, I \to W(R))$$

consisting of two generalized Cartier divisors and a Cartier-Witt divisor, together with an isomorphism

$$(F_*I \otimes_{F_*W(R)} \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R) \to \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R)) \simeq (\mathcal{L}_1 \otimes_R \mathcal{L}_2 \to \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R))$$

of generalized Cartier divisors.

3.3.11 (**The Nygaard filtered Prismatisation**)**.** Out of the data of a filtered Cartier-Witt divisor, we can construct a Cartesian square of filtered animated rings

$$\begin{array}{ccc} Fil^{\bullet}_{M} W(R) & \longrightarrow & Fil^{\bullet}_{F_* I} F_* W(R) \\ \downarrow & & \downarrow \\ Fil^{\bullet}_{\mathcal{L}_1} R & \longrightarrow & Fil^{\bullet}_{\mathcal{L}_1} \mathbf{G}_{\mathrm{a}}^{\mathrm{dR}}(R) \end{array}$$

where the lower horizontal map is the canonical one and the right vertical map uses the identification in the data of a filtered Cartier-Witt divisor [Hau25][2.3.12]. Thus we obtain a (co)fiber sequence

$$M(R) \to W(R) \to \mathbf{G}_{\mathrm{a}}^{\mathcal{N}}(R)$$

with the cofibre carrying the structure of an animated ring. This defines a ring stack

$$\mathbf{G}_{\mathrm{a}}^{\mathcal{N}} \to \mathbf{Z}_p^{\mathcal{N}}$$

over the moduli of filtered Cartier-Witt divisors. Finally for a p-adic formal scheme X, we define the *Nygaard filtered Prismatisation* via transmutation, by assigning a Nygaard filtered Cartier-Witt divisor to the anima

$$X^{\mathcal{N}}(\mathrm{Spec}(R) \to \mathbf{Z}_p^{\mathcal{N}}) := X(\mathrm{Spec}(\mathbf{G}_{\mathrm{a}}^{\mathcal{N}}(\mathrm{Spec}(R) \to \mathbf{Z}_p^{\mathcal{N}}))).$$

3.3.12. The Prismatisation sits openly inside the Nygaard filtered prismatisation in two disjoint ways

$$X^{\mathbb{\Delta}} \xrightarrow{j_{dR}} X^{\mathcal{N}} \xleftarrow{j_{HT}} X^{\mathbb{\Delta}}.$$

Here the open immersion on the left comes from pulling back along $\mathbf{G}_{\mathrm{m}} / \mathbf{G}_{\mathrm{m}} \to \mathbf{A}^1 / \mathbf{G}_{\mathrm{m}}$ and the open immersion on the right comes from pulling back along $(\mathbf{G}_{\mathrm{m}} / \mathbf{G}_{\mathrm{m}})^{\mathrm{dR}} \to (\mathbf{A}^1 / \mathbf{G}_{\mathrm{m}})^{\mathrm{dR}}$. Here composing j_{dR} with the structure map recovers the identity, while composing j_{HT} with the structure map recovers the Frobenius.

Example 3.3.13. Given a filtered animated ring $Fil^{\bullet} A$, we can associate to it the so called *Rees stack*

$$\mathcal{R}(Fil^{\bullet} A) := \mathrm{Spec}(\oplus_{i \in \mathbf{Z}} Fil^i A \cdot t^{-1}) / \mathbf{G}_{\mathrm{m}} \to \mathbf{A}^1 / \mathbf{G}_{\mathrm{m}}$$

where the $\mathbf{G}_{\mathrm{m}}$-action comes from the grading. Then one obtains an equivalence

$$\mathcal{D}(\mathcal{R}(Fil^{\bullet} A)) \simeq \mathrm{Mod}_{Fil^{\bullet} A}(\mathcal{DF}(A))$$

where on the right we denote by $\mathcal{DF}(A)$ the category of filtered objects in $\mathcal{D}(A)$ [Mou19].

Now given a semiperfectoid S, by [BL22b][5.5.10] there is a canonical identification

$$\mathcal{R}(Fil^{\bullet}_{\mathcal{N}} \mathbb{\Delta}_S) \simeq R^{\mathcal{N}}$$

where we write $Fil^{\bullet}_{\mathcal{N}} \mathbb{\Delta}_S$ for the absolute Nygaard filtration [BL22a][5.5].

Example 3.3.14. Continuing 3.3.13, let us assume R is integral perfectoid and $A \simeq \mathbb{A}_{\mathrm{inf}}(R)$. Then we have an isomorphism of graded rings

$$\oplus_{i \in \mathbf{Z}} Fil^i_{\mathcal{N}} A \cdot t^{-1} \simeq A[t,u]/(tu - \varphi^{-1}(d))$$

where d is a generator of I and $u = \varphi^{-1}(d) t^{-1}$ sits in degree -1.

From this description, we obtain the following Zariski stratification:

(a) The locus $(R^{\mathcal{N}})_{t\neq 0}$ corresponds to taking the colimit of the filtration and thus recovers $\mathrm{Spf}(A) \simeq R^{\mathbb{\Delta}}$.
(b) The locus $(R^{\mathcal{N}})_{t=0,u\neq 0}$ identifies with $\mathrm{Spf}(A/\varphi^{-1}(d)) \simeq \mathrm{Spf}(\varphi_* R)$ via the filtered Frobenius [BL22b][5.5.6].
(c) The locus $(R^{\mathcal{N}})_{u=t=0}$ identifies with $\mathrm{B}\mathbf{G}_{\mathrm{m}}$ over $\mathrm{Spf}(\varphi_* R)$.

Example 3.3.15. We further continue on 3.3.14 and describe the ring stack $\mathbf{G}_{\mathrm{a}}^{\mathcal{N}}$ over a perfectoid R. That is, in order to describe the pullback of the latter ring stack to $R^{\mathcal{N}}$, we can consider the following diagram of A-module schemes

$$\begin{array}{ccccc}
\varphi^{-1}(I) \otimes_A \mathbf{G}_{\mathrm{a}}^{\#} & \longrightarrow & \varphi^{-1}(I) \otimes_A W & \longrightarrow & F_*(I \otimes_A W) \\
\downarrow{\scriptstyle u^\sharp} & & \downarrow & & \| \\
\mathbf{V}(\mathcal{L}_2)^\sharp & \longrightarrow & M & \longrightarrow & F_*(I \otimes_A W) \\
\downarrow{\scriptstyle t^\sharp} & & \downarrow & & \downarrow \\
\mathbf{G}_{\mathrm{a}}^{\#} & \longrightarrow & W & \longrightarrow & F_* W
\end{array}$$

where the horizontal compositions are fiber sequences and the vertical compositions are the canonical maps. Then we obtain

$$W/M \simeq \mathbf{G}_{\mathrm{a}}^{\mathcal{N}}.$$

Thus, further restricting to the stratification described in 3.3.14 we obtain:

(a) Over $(R^{\mathcal{N}})_{t\neq 0}$ we obtain
$$\mathbf{G}_{\mathrm{a}}^{\mathcal{N}} \simeq F_*(W/I) \simeq \varphi^* \mathbf{G}_{\mathrm{a}}^{\mathbb{\Delta}}$$
as one sees by taking cofibres of the lower vertical maps.
(b) Over $(R^{\mathcal{N}})_{t=0,u\neq 0}$, we obtain
$$\mathbf{G}_{\mathrm{a}}^{\mathcal{N}} \simeq W/\varphi^{-1}(I) \simeq \varphi^* \mathbf{G}_{\mathrm{a}}^{\mathrm{HT}}.$$
(c) Over $(R^{\mathcal{N}})_{u=t=0}$ we first observe that
$$M \simeq \mathbf{V}(\mathcal{L}_2)^\sharp \oplus F_*(I \otimes_A W).$$
Furthermore, as we live over the Hodge-Tate locus, the Cartier-Witt divisor is locally given by $V(1)\colon W \to W$. Such that via this identification the map $M \to W$ sits in a pushout square
$$\begin{array}{ccc}
\mathbf{V}(\mathcal{L}_2)^\sharp \oplus F_* W & \xrightarrow{(0,V)} & W \\
\downarrow{\scriptstyle pr} & & \downarrow{\scriptstyle pr} \\
\mathbf{V}(\mathcal{L}_2)^\sharp & \xrightarrow[0]{} & \mathbf{G}_{\mathrm{a}}.
\end{array}$$
From this we learn that over this locus we have an identification
$$\mathbf{G}_{\mathrm{a}}^{\mathcal{N}} \simeq \mathbf{G}_{\mathrm{a}} \oplus B\mathbf{V}(\mathcal{L}_2)^\sharp \simeq \mathbf{G}_{\mathrm{a}}^{\mathrm{Hodge}}.$$

Finally, unwinding what we did for a map $X \to \mathrm{Spf}(R)$ of p-adic formal schemes, we learn that we obtain a stratification:

(a) $(X^{\mathcal{N}})_{t\neq 0} \simeq X^{\mathbb{\Delta}}$ over $\mathrm{Spf}(A)$.
(b) $(X^{\mathcal{N}})_{t=0,u\neq 0} \simeq \varphi_* X^{\mathrm{HT}}$ over $\mathrm{Spf}(\varphi_* R)$.
(c) $(X^{\mathcal{N}})_{u=t=0} \simeq \varphi_* X^{\mathrm{Hodge}}$ over $\mathrm{B}\mathbf{G}_{\mathrm{mSpf}(\varphi_* R)}$.

3.4. **Recollections on the Hodge-Tate locus.** One can understand the geometry of the Hodge-Tate stack 3.3.8 quite well. In order to make use of this later, we need to recall some aspects of these stacks. For this, we will essentially copy [BL22c][5], but taking care that we work in a slightly different topology.

3.4.1. Let us write

$$\mathbf{G}_{\mathrm{a}}^{\#}\{1\} \in (\mathbf{Z}_p^{HT})_{\text{éid}}$$

for the sheaf which takes a Hodge-Tate divisor $I \to W(S)$ on $\mathrm{Spf}(S)$ to $\mathbf{G}_{\mathrm{a}}^{\#}(S)\{1\}$ where the Breuil-Kisin twist was defined in 3.3.9. This defines a group object and we will write $\mathrm{B}\mathbf{G}_{\mathrm{a}}^{\#}\{1\}$ for the sheaf of torsors on it. Not that using 3.2.6 and 3.2.5 we obtain an identification

$$\mathrm{B}\mathbf{G}_{\mathrm{a}}^{\#}\{1\} \simeq R\Gamma_{\text{éid}}(_, W[F]\{1\}[1])$$

where the twist $\{1\}$ on the right is defined analogously.

Let us write $\mathbf{G}_{\mathrm{a}}^{\mathrm{HT}}$ for the ring stack over $\mathbf{Z}_p^{HT}$ obtained by pulling back $\mathbf{G}_{\mathrm{a}}^{\mathbb{\Delta}}$. The following, then, is an absolute version of [BL22c][5.10] with the same proof.

Proposition 3.4.2. *The ring stacks $\mathbf{G}_{\mathrm{a}}^{\mathrm{HT}}$ is a square-zero extension of $\mathbf{G}_{\mathrm{a}}$ by $\mathrm{B}\mathbf{G}_{\mathrm{a}}^{\#}\{1\}[-1]$. That is there exists a natural W-linear derivation $\partial\colon \mathbf{G}_{\mathrm{a}} \to \mathbf{G}_{\mathrm{a}} \oplus \mathrm{B}\mathbf{G}_{\mathrm{a}}^{\#}\{1\}$ fitting into a cartesian square*

$$\begin{array}{ccc} \mathbf{G}_{\mathrm{a}}^{\mathrm{HT}} & \longrightarrow & \mathbf{G}_{\mathrm{a}} \\ {\scriptstyle \pi^{HT}}\downarrow & & \downarrow{\scriptstyle \partial} \\ \mathbf{G}_{\mathrm{a}} & \xrightarrow[\partial_{triv}]{} & \mathbf{G}_{\mathrm{a}} \oplus \mathrm{B}\mathbf{G}_{\mathrm{a}}^{\#}\{1\} \end{array}$$

of ring stacks over $\mathbf{Z}_p^{HT}$.

Proof. First, all ring stacks in the square define sheaves for the éid-topology. For the lower right corner, this follows from the identification 3.4.1, for the left upper corner from 3.3.4, and for $\mathbf{G}_{\mathrm{a}}$ as the topology is subcanonical. That means to show the claim for the values on an animated Hodge-Tate divisor $\alpha\colon I \to W(S)$, we can resolve $\mathbf{Z}_p^{HT}$ by (ordinary) prisms and show the claim on the induced cover of $\mathrm{Spf}(S)$. Concretely, we can, for example, take the covering from 3.3.6 then all objects appearing in the Čech nerve are represented by (ordinary) prisms [BL22a][3.2.8+3.2.10]. This reduces the claim to the situation relative to a prism, in which case the argument is given in [BL22c][5.10]. For the convenience of the reader, we recall this argument now.

We fix a prism (A, I) and let us assume that p is a non-zero divisor in $\overline{A}$. Using 3.4.1 we can replace $\mathrm{B}\mathbf{G}_{\mathrm{a}}^{\#}\{1\}$ by $R\Gamma(_, W[F]\{1\}[1])$ and by taking limits it suffices to prove the corresponding claim for $R\Gamma(_, W_n[F]\{1\}[1])$ functorial in n.

We first prove this claim evaluated at a static p-nilpotent $\overline{A}$-algebra S for which the Frobenius maps

$$F\colon W_n(S) \to F_* W_{n-1}(S)$$

are surjective for all $n \geq 2$. To see this, we first claim:

(∗) The map $\alpha\colon I \otimes_{W(S)} W_n(S) \to W_n(S)$ maps surjectivity onto $VW_n(S)$.

Using this claim, we can compute

$$\pi_0\overline{W_n}(S) \simeq W_n(S)/VW_n(S) \simeq S$$

and

$$\pi_1\overline{W_n}(S) \simeq ker(I \otimes_{W(S)} W_n(S) \to W_n(S)) \simeq I \otimes_{W(S)} W_n[F](S).$$

So the claim of the proposition, in this case, follows as any 1-truncated animated ring is naturally a square zero extension of its π_0 by its $\pi_1[1]$. To see $(*)$ we can work zariski locally on $W(S)$ and thus assume α corresponds to a distinguished element $d = (x_0, x_1, \dots) \in W(S)$. The assumption that the Cartier-Witt divisor lives in the Hodge-Tate stack tells us that $x_0 = 0$, such that $d = V(u)$ where u is a unit as d is distinguished. Thus the formula $V(u) \cdot \underline{x} = V(u \cdot F(\underline{x}))$ shows that α maps into the image of the Verschiebung map. On the other hand, as the Frobenius is surjective, we can write any $\underline{y} \in W_{n+1}(S)$ as $\underline{y} = u \cdot F(\underline{x})$ for some $\underline{x}$ and the surjectivity follows from the equality

$$V(\underline{y}) = V(u \cdot F(\underline{x})) = V(u) \cdot \underline{x}.$$

To deduce the claim of the proposition, we proceed as follows. As all functors are sheaves for the éid-topology, by 3.2.8, we can descend the statement to all static p-nilpotent $\overline{A}$ algebras. This in particular implies the statement for all polynomial algebras over $\overline{A}/p^n$. Now all functors are left Kan extended from polynomial algebras, such that we obtain the claim for all p-nilpotent animated $\overline{A}$-algebras. The general claim now follows by left Kan extending from affine p-nilpotent schemes. □

3.4.3. For any morphism $f\colon X \to S$ of p-adic formal schemes precomposing with the map $\pi_{HT}\colon \mathbf{G}_{\mathrm{a}}^{\mathrm{HT}} \to \mathbf{G}_{\mathrm{a}}$ induces a map

$$\pi_f^{HT}\colon X^{HT} \to X \times_S S^{HT}$$

which we will call the *Hodge-Tate structure map* of f. Using 3.4.2, we can understand the geometry of this map in terms of the cotangent complex of f.

3.4.4. Given a square zero extension $\tilde{B} \to B$ along a connective B-module N. Then, for any morphism $X \to S$ of (derived formal) schemes and any point $\eta \in X(B) \times_{S(B)} S(\tilde{B})$ the fibre at η of the map

$$X(\tilde{B}) \to X(B) \times_{S(B)} S(\tilde{B})$$

gives a torsor over $\mathrm{Hom}_X(\mathcal{L}_f, N)$. Let us informally[9] describe the action. The point η corresponds to a commutative square like the outer square in the following diagram

$$\begin{array}{ccccc} \mathcal{O}_S & \longrightarrow & \mathcal{O}_X \times_B \tilde{B} & \longrightarrow & \tilde{B} \\ \downarrow & \nearrow & \downarrow & \nearrow & \downarrow \\ \mathcal{O}_X & \longrightarrow & \mathcal{O}_X & \longrightarrow & B. \end{array}$$

Thus, the fiber is given by the anima of dashed lifts in the square. This anima is equivalent to the dashed lifts in the left square, but now choosing such a lift gives an identification of $\mathcal{O}_X \times_B \tilde{B}$ with the trivial square zero extension. Via this identification, the anima of lifts, by definition, becomes the anima of $\mathcal{O}_S$-linear derivations in N, which is isomorphic to $\mathrm{Hom}_X(\mathcal{L}_f, N)$.

[9] To make this coherent, we have to specify a point in $\mathrm{BHom}_X(\mathcal{L}_f, N) \simeq \mathrm{Hom}_X(\mathcal{L}_f, N[-1])$. But the trivial map does the job.

Notation 3.4.5. Consider a morphism $f\colon X \to S$ of p-adic formal schemes. Then, we will write

$$\mathbf{V}(\mathcal{L}_f\{1\})^\sharp \to X \times_S S^{HT}$$

for the bundle which takes a point $\mathrm{Spec}(R) \to X \times_S S^{HT}$ to

$$\mathrm{Hom}_R(\eta_X^* \mathcal{L}_f, \mathbf{G}_{\mathrm{a}}^\#\{1\}(R))$$

where η corresponds to the point in X and the Breuil-Kisin twist is defined via the point in S^{HT}. Note that this defines a group object. The following proposition is just a reformulation of [BL22c][5.12] in its natural generality.

Proposition 3.4.6. *Given a map $f\colon X \to S$ of derived p-adic formal schemes, the associated Hodge-Tate structure map*

$$\pi_f^{HT}\colon X^{HT} \to X \times_S S^{HT}$$

defined a gerbe baned by $\mathbf{V}(\mathcal{L}_f\{1\})^\sharp$.

Proof. We claim that for any point $\eta\colon \mathrm{Spec}(R) \to X \times_S S^{HT}$ the fibre of the map

$$X^{HT}(R) \to X(R) \times_{S(R)} S^{HT}(R)$$

at η defines a torsor over

$$B\mathbf{V}(\mathcal{L}_f)^\sharp(R) \simeq \mathrm{Hom}_R(\eta_X^* \mathcal{L}_f, \mathrm{B}\mathbf{G}_{\mathrm{a}}^\#(R)).$$

This follows by combining 3.4.4 and 3.4.2. □

Corollary 3.4.7. *For any map $f\colon X \to S$ of p-adic formal schemes, which has vanishing cotangent complex the square*

$$\begin{array}{ccc} X^{HT} & \longrightarrow & S^{HT} \\ {\scriptstyle \pi^{HT}}\downarrow & & \downarrow{\scriptstyle \pi^{HT}} \\ X & \xrightarrow[f]{} & S \end{array}$$

is Cartesian.

In the following statement, we will make use of the notion of derived algebras in the sense of [MM25].

Proposition 3.4.8. *Consider an affine p-adic formal scheme $f\colon Spf(B) \to Spf(\mathbf{Z}_p)$, then the map*

$$B^{Hodge} \to \mathrm{B}\mathbf{G}_{\mathrm{m}}$$

is relatively affine and represented by the graded (possibly non-connective) derived algebra

$$\oplus_i \Gamma(Spf(B), \wedge^i \mathcal{L}_f[-i])(i)$$

computing absolute Hodge cohomology.

Proof. Note that both sides are compatible with colimits in B. For B^{Hodge}, this follows as it is defined by transmutation, and for the absolute Hodge cohomology this follows as it is left Kan-extended from polynomial algebras. Thus it suffices to produce the identification for polynomial algebras. But for polynomial algebras, we can combine 3.4.6 and [MM25][4.19] to obtain the affineness. The computation of the global sections in this case follows from Cartier duality (see [BL22b][2.5.6]). □

3.5. **Solid Hodge(-Tate) stacks.** In the following section we will work with the topos $\mathrm{Spf}(\mathbf{Z}_p)_{\text{éid}}$ from 2.2.7. Recall from 2.3.3 that there is a functor

$$(_)^{\square}\colon \mathrm{Spf}(\mathbf{Z}_p)_{\text{éid}} \to \mathrm{AnStack}_{/\mathrm{Spa}(\mathbf{Z}_p)}$$

which preserves finite limits and colimits.

3.5.1. Given a p-adic formal stack $X \in \mathrm{Spf}(\mathbf{Z}_p)_{\text{éid}}$ and a locally connective quasi coherent sheaf $\mathcal{E} \in \mathcal{D}(X)$, we consider the bundle

$$\mathbf{V}(\mathcal{E})^{\sharp} \to X.$$

Concretely this bundle sends a point $\eta_A\colon \mathrm{Spf}(A) \to X$ to the anima

$$\mathrm{Hom}_{\mathcal{D}(\mathrm{Spf}(A))}(\eta_A^*\mathcal{E}, \mathbf{G}_{\mathrm{a}}^{\#}(A)) = \mathrm{Hom}_{\mathcal{D}(\mathrm{Spf}(A))}(\Gamma(\eta_A^*\mathcal{E}), A)$$

and thus defines a group object in $X_{\text{éid}}$.

The goal of this section is now to prove the following.

Theorem 3.5.2. *Given a p-adic formal stack $X \in Spf(\mathbf{Z}_p)_{éid}$ together with a locally connective perfect complex $\mathcal{E} \in Perf(X)$, we have the following.*

(1) For any $\mathbf{V}(\mathcal{E})^{\sharp}$-gerbe $G \to X$, the induced map

$$G^{\square} \to X^{\square}$$

is weakly proper.

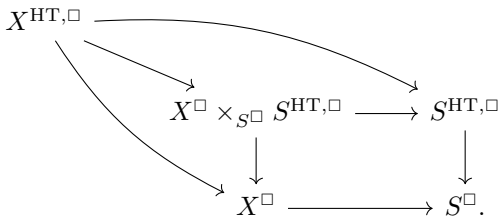
Before proving this theorem, let us record some corollaries.

Corollary 3.5.3. *Consider a proper map of p-adic formal schemes $X \to S$. Then the induced map*

$$X^{HT,\square} \to S^{HT,\square}$$

is weakly proper.

Proof. Consider the commutative diagram



The lower horizontal map is weakly proper by 2.3.4, so the upper horizontal map is also weakly proper. By 3.4.6, the map into the fiber product is a $\mathbf{V}(\mathcal{L}_{X/S}\{1\})^{\sharp}$-gerbe and thus weakly proper by 3.5.2 (note that the cotangent complex is a perfect complex because the map is of finite presentation). This shows what we want, as weakly proper maps are stable under composition. □

Corollary 3.5.4. *Consider a proper map $X \to S$ of p-adic formal schemes. Then the induced map*

$$X^{Hodge,\square} \to S^{Hodge,\square}$$

is weakly proper.

PROOF. Note that $\mathbf{G}_{\mathrm{a}}^{\mathrm{Hodge}} \to \mathbf{G}_{\mathrm{a}}$ is a square zero extension by $B\mathbf{V}(\mathcal{O}(-1))^\sharp$ of ring stacks over $\mathrm{B}\mathbf{G}_{\mathrm{m}}$. Using the same argument as in 3.4.6 we see that the map

$$X^{\mathrm{Hodge}} \to X \times_S S^{\mathrm{Hodge}}$$

is a $\mathbf{V}(\mathcal{L}_{X/S}(-1))^\sharp$-gerbe. So the claim follows the same way as in 3.5.3. □

Let us now come to the proof of 3.5.2.

Construction 3.5.5. Let (for simplicity) R be a static ring. Then for a free R-module $\mathcal{E}$, we consider the divided power deRham complex

$$R \longrightarrow \Gamma_R(\mathcal{E}) \longrightarrow \Gamma_R(\mathcal{E}) \otimes M \xrightarrow{d} \Gamma_R(\mathcal{E}) \otimes \wedge^2 M \xrightarrow{d} \cdots.$$

By the divided power Poincaré lemma [Sta, Tag 07LC] this chain complex is acyclic and thus gives an isomorphism

$$R \simeq \mathrm{dR}^{\mathrm{dp}}_{B/R}$$

where the right-hand side denotes the complex starting at the first stage and B the free algebra on $\mathcal{E}$. As a chain complex the object $\mathrm{dR}^{\mathrm{pd}}_{B/R}$ corresponds to a filtration

$$Fil^\bullet_{\mathrm{Hodge}} \mathrm{dR}^{\mathrm{dp}}_{B/R} \to R$$

which we will refer to as the *Hodge filtration*. Animating this construction, we thus obtain a Hodge filtration resolving R for any connective object $\mathcal{E} \in \mathcal{D}(R)$. The graded pieces of this filtration are given by

$$gr^i_{\mathrm{Hodge}} \mathrm{dR}^{\mathrm{dp}}_{B/R} \simeq \wedge^i \Gamma_R(\mathcal{E})[-i].$$

Note also that this filtration lives in $\Gamma_R(\mathcal{E})$-comodules.

We need the following lemma.

Lemma 3.5.6. Consider an animated ring R and a connective perfect complex $\mathcal{E} \in \mathcal{D}(R)$. Furthermore, let B denote the free R-algebra on $\mathcal{E}$. Then the Hodge filtration

$$Fil^\bullet_{\mathrm{Hodge}} \mathrm{dR}^{\mathrm{dp}}_{B/R}$$

is finite.

PROOF. By construction, the functor

$$Fil^\bullet_{\mathrm{Hodge}} \mathrm{dR}^{\mathrm{dp}}_{B(_)/R} \colon \mathcal{D}(R)_{\geq 0} \to \mathrm{CAlg}(\mathcal{DF}(R))$$

preserves colimits. As on the right, finite filtrations are stable under finite colimits and retractions, it is enough to observe the claim for $\mathcal{E} \simeq R$. This is clear. □

Proof of 3.5.2(1). By 1.1.11[c] we can assume that $X = \mathrm{Spf}(A)$ is affine and $G \simeq B\mathbf{V}(\mathcal{E})^\sharp_X$ for some connective perfect complex $\mathcal{E}$ on X. We now claim that the map $X \to G$ is a descendable cover. For this first recall that we can identify $\mathcal{D}(G)$ with $\Gamma_R(\mathcal{E})$-comodules in $\mathcal{D}(X)$. So to see the descendability, we can use the filtration from 3.5.5, which is finite by 3.5.6 and has graded pieces given by perfect complexes.

In the following, we will in particular deduce tha the diagonal is weakly cohomologically proper. So by 1.1.12 and 1.1.11[c], it suffices to see that the map $X^\square \to G^\square$ is co-smooth. This we can see locally on G, so it reduces to checking that the map

$$(\mathbf{V}(\mathcal{E})^\sharp_X)^\square \to X^\square$$

is co-smooth. By 2.3.3[D], we can check this after base changing to $\mathrm{Spec}(\mathbf{F}_p)$. Then the map in question becomes

$$\mathrm{Spa}(\Gamma_{A/p}(\mathcal{E}/p)) \to \mathrm{Spa}(A/p)$$

which is a map of affine discrete adic spaces. As $\mathcal{E}/p$ is built out of colimits from A/p, the functor $\Gamma_{A/p}(_)$ commutes with colimits and proper maps of analytic rings are stable under colimits, we see that it suffices to check that

$$(\mathbf{G}_{\mathrm{a}}^{\#})_{A/p} \to \mathrm{Spec}(A/p)$$

becomes proper. That this map is integral can either be seen by hand or follows from [BL22b][2.6.5]. □

From the proof we also obtain the following corollary.

Proposition 3.5.7. *Consider a map $f\colon Spf(S) \to Spf(R)$ and assume one of the following:*

(a) R is semiperfectoid and f is a naive syntomic cover.
(b) R is semiperfectoid and f is a zariski closed immersion.

then the induced map

$$S^{Hodge,\square} \to R^{Hodge,\square}$$

is weakly proper.

Proof. We start with (a). As weakly proper maps are stable under base change and composition, we can assume that f is a generic naive syntomic cover. Furthermore, by choosing a perfectoid surjecting onto R and lifting the cover, we can assume R to be perfectoid. Now we consider the cofiber sequence

$$f^* \mathcal{L}_{\mathrm{Spf}(R)} \to \mathcal{L}_{\mathrm{Spf}(S)} \to \mathcal{L}_f$$

where the left-hand side is a free module sitting in homological degree 1 as R is perfectoid and the right-hand side is free sitting in homological degree 1 by 3.1.5. So the same also holds for the middle term. Using 3.4.8 we thus learn that

$$S^{\mathrm{Hodge},\square} \to R^{\mathrm{Hodge},\square}$$

is an affine map of analytic stacks. As proper maps of analytic rings are stable under colimits (and p-adic completion), we now can assume that the set of generators for the generic naive syntomic cover $\mathrm{Spf}(S) \to \mathrm{Spf}(R)$ is finite, such that $\mathcal{L}_f$ becomes finite free sitting in homological degree 1. Finally using 2.3.3[D], we can check the weakly properness modulo p and the last part of the proof of 3.5.2 shows that $S^{\mathrm{Hodge},\square} \times_{\mathrm{Spa}(\mathbf{Z}_p)} \mathrm{Spa}(\mathbf{F}_p)$ lives in the subtopos of

$$\mathrm{AnStack}_{/R^{\mathrm{Hodge},\square} \times_{\mathrm{Spa}(\mathbf{Z}_p)} \mathrm{Spa}(\mathbf{F}_p)}$$

generated by proper analytic rings over the base. Any affine map in there is weakly proper. To prove (b), by a similar argument as above, we first observe that the cotangent complex of a semiperfectoid is 1-connective. As both R and S are semiperfectoid, using 3.4.8, we see that the map

$$S^{\mathrm{Hodge},\square} \to R^{\mathrm{Hodge},\square}$$

is an affine map of analytic stacks. Now writing S as a colimit of finite presentation Zariski quotients of R and using that proper maps of analytic rings are stable under colimits, we deduce the claim from 3.5.4. □

3.6. **Inducing covers on Prismatic stacks.** In this section we will explain how to induce covers on the Nygaard filtered prismatisation.

3.6.1. In the following we will make use of the functor

$$(_)^{\square}\colon (\mathbf{Z}_p^{\mathbb{\Delta}})_{\text{éid}} \to \mathrm{AnStack}_{/\mathbf{Z}_p^{\mathbb{\Delta},\square}}$$

introduced in 2.3.3[A]. Note that using 3.3.13, we see that for a semiperfectoid R, the stack $R^{\mathcal{N}}$ lives in the domain of this functor.

The main theorem of this section is the following.

Theorem 3.6.2. *Consider a map $g\colon Spf(S) \to Spf(R)$ of affine p-adic formal schemes and assume $Spf(R)$ is semiperfectoid. Then we have the following:*

(a) If f is a Zariski open immersion (resp. a Zariski cover), then the induced map
$$S^{\mathcal{N},\square} \to R^{\mathcal{N},\square}$$
is an open immersion (resp. an open covering) of analytic stacks.

(b) If f is étale (resp. an étale covering), then the induced map
$$S^{\mathcal{N},\square} \to R^{\mathcal{N},\square}$$
is an étale map (resp. étale cover) of analytic stacks.

(c) If f is an infinite p-root cover, the induced map
$$S^{\mathcal{N},\square} \to R^{\mathcal{N},\square}$$
is a proper descendable cover of analytic stacks.

Before proving the theorem let us also state the following corollary.

Corollary 3.6.3. *Consider a map $g\colon Spf(S) \to Spf(R)$ of affine p-adic formal schemes and assume $Spf(R)$ is semiperfectoid. Then we have the following:*

(a) If f is a Zariski open immersion (resp. a Zariski cover), then the induced map
$$S^{\mathbb{\Delta},\square} \to R^{\mathbb{\Delta},\square}$$
is an open immersion (resp. an open covering) of analytic stacks.

(b) If f is étale (resp. an étale covering), then the induced map
$$S^{\mathbb{\Delta},\square} \to R^{\mathbb{\Delta},\square}$$
is an étale map (resp. étale cover) of analytic stacks.

(c) If f is an infinite p-root cover, the induced map
$$S^{\mathbb{\Delta},\square} \to R^{\mathbb{\Delta},\square}$$
is a proper descendable cover of analytic stacks.

Proof. This follows from 3.6.2 and the cartesian square

$$\begin{array}{ccc} S^{\mathbb{\Delta},\square} & \longrightarrow & S^{\mathcal{N},\square} \\ \downarrow & & \downarrow \\ R^{\mathbb{\Delta},\square} & \longrightarrow & R^{\mathcal{N},\square} \end{array}$$

using the open immersion j_{dR} from 3.3.12. □

Proof of 3.6.2(a) and (b). We start with the case of an elementary Zariski open immersion (resp. a Zariski cover consisting of elementary open immersions). Choosing a perfectoid mapping surjective to R, we can lift the open immersion (resp. the cover) to this perfectoid, and as the conclusion is stable under base change, we can assume R is perfectoid. In this case S is perfectoid as well, and we have to check that the map

$$\mathrm{Spa}(\mathbb{A}_{\mathrm{inf}}(S)[t,u]/(tu-\varphi^{-1}(d))) \to \mathrm{Spa}(\mathbb{A}_{\mathrm{inf}}(R)[t,u]/(tu-\varphi^{-1}(d)))$$

is an open immersion (resp. an open cover). Using 2.3.3[B], we see that it is enough to show that the map

$$\mathrm{Spf}(\mathbb{A}_{\mathrm{inf}}(S)) \to \mathrm{Spf}(\mathbb{A}_{\mathrm{inf}}(R))$$

is an open immersion (resp. a Zariski cover). This follows from deformation theory, as the conclusion holds for the map $\mathrm{Spf}(S) \to \mathrm{Spf}(R)$.

To see the assertion in (b), we can now work Zariski locally on R. Locally on R the map $\mathrm{Spf}(S) \to \mathrm{Spf}(R)$ factors as an open immersion followed by a standard étale map (or a collection of such maps in the case of an étale cover). As above, such a map can be lifted to a perfectoid, so we may assume R is perfectoid. Again S is perfectoid in this case as well, and we argue as in (a) to deduce the claim. □

In order to prove part (c), we will need some preparations. First, let us recall some facts on filtered complexes.

Definition 3.6.4. Given an animated ring R, we will write

$$\mathcal{DF}(R) := \mathrm{Fun}(\mathbf{Z}^{\mathrm{op}}, \mathcal{D}(R))$$

for the category of *filtered objects* in $\mathcal{D}(R)$. Here we understand $\mathbf{Z}^{\mathrm{op}}$ as a poset.

Remark 3.6.5. We often work with the full subcategory of derived I-complete objects in $\mathcal{DF}(R)$ for some ideal in R. But it will be obvious how to adopt everything we say about this category.

3.6.6. Taking the colimit and taking the cofiber of each map in the filtration induces functors

$$\mathcal{D}(R) \xleftarrow{\mathrm{colim}} \mathcal{DF}(R) \xrightarrow{\mathrm{gr}^\bullet} \mathrm{Fun}(\mathbf{Z}, \mathcal{D}(R))$$

where we understand $\mathbf{Z}$ as a discrete category. We will refer to the category $\mathrm{Fun}(\mathbf{Z}, \mathcal{D}(R))$ as *graded objects* in $\mathcal{D}(R)$.

3.6.7. Given two filtered objects $\mathcal{F}^\bullet, \mathcal{E}^\bullet \in \mathcal{DF}(R)$, the mapping spectrum

$$\mathrm{Hom}_{\mathcal{DF}(R)}(\mathcal{F}^\bullet, \mathcal{E}^\bullet)$$

can be computed as the equalizer of the two maps

$$\prod_{i\in\mathbf{Z}^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{D}(R)}(\mathcal{F}^i, \mathcal{E}^i) \rightrightarrows \prod_{j\in\mathbf{Z}^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{D}(R)}(\mathcal{F}^j, \mathcal{E}^{j-1})$$

one coming from postcomposing along the target filtration and one from precomposing along the source filtration. This follows as one can write the category $\mathbf{Z}^{\mathrm{op}}$ as the Segal completion of the simplicial anima

$$\ldots \Delta^1 \coprod_{\Delta^0} \Delta^1 \coprod_{\Delta^0} \Delta^1 \ldots.$$

3.6.8. Day-convolution equips the category $\mathcal{DF}(R)$ with a symmetric monoidal structure. That is, we obtain the formula

$$(\mathcal{F}^\bullet \otimes \mathcal{E}^\bullet)^n \simeq \mathrm{colim}_{i+j\geq n}\, \mathcal{F}^i \otimes \mathcal{E}^j.$$

Furthermore, this monoidal structure is closed, so we have an internal hom. Note also that the colimit functor from 3.6.6 is symmetric monoidal.

Day-convolution also equips the category $\mathcal{D}(R)_{\mathrm{graded}}$ of graded objects with a symmetric monoidal structure. For two graded objects $\mathcal{F}^\bullet, \mathcal{E}^\bullet$ this produces the formula

$$(\mathcal{F}^\bullet \otimes \mathcal{E}^\bullet)^n \simeq \mathrm{colim}_{i+j=n}\, \mathcal{F}^i \otimes \mathcal{E}^j \simeq \bigoplus_{j+i=n} \mathcal{F}^i \otimes \mathcal{E}^j.$$

Furthermore, the associated graded functor from 3.6.6 is symmetric monoidal for these structures. Also, the symmetric monoidal structure on graded objects is closed. Unwinding the formula for the internal hom, one gets

$$\underline{\mathrm{Hom}}_{\mathcal{D}(R)_{\mathrm{graded}}}(\mathcal{F}^\bullet, \mathcal{E}^\bullet)^n \simeq \prod_{m\in\mathbf{Z}} \underline{\mathrm{Hom}}_{\mathcal{D}(R)}(\mathcal{F}^m, \mathcal{E}^{m-n}).$$

The associated graded functor is also compatible with the internal homs [GP16][2.28], that is, we have the formula

$$\mathrm{gr}^\bullet \underline{\mathrm{Hom}}_{\mathcal{DF}(R)}(\mathcal{F}^\bullet, \mathcal{E}^\bullet) \simeq \underline{\mathrm{Hom}}_{\mathcal{D}(R)_{\mathrm{graded}}}(\mathrm{gr}(\mathcal{F}^\bullet), \mathrm{gr}(\mathcal{E}^\bullet)).$$

3.6.9. A filtered object $\mathcal{F}^\bullet$ in $\mathcal{D}(R)$ is called complete, if

$$\lim_{n\in\mathbf{Z}^{\mathrm{op}}} \mathcal{F}^n \simeq 0.$$

The full subcategory $\widehat{\mathcal{DF}}(R) \subset \mathcal{DF}(R)$ of complete filtered objects admits a symmetric monoidal left adjoint [GP16][2.25]. Furthermore, the associated grade functor is conservative when restricted to complete objects.

3.6.10. There is a t-structure on $\mathcal{DF}(R)$, called the *standard t-structure*. The connective objects are given by those filtered objects $\mathcal{F}^\bullet$, for which each $\mathcal{F}^i$ is connective in the standard t-structure on $\mathcal{D}(R)$.

The crucial input will be the following. Recall that for an algebra $R \to S$ over an integral perfectoid with corresponding perfect prism A, we write $Fil^\bullet_{\mathcal{N}} F^* \mathbb{\Delta}_{S/A}$ for the relative Nygaard filtration [BL22a][5.1].

Proposition 3.6.11. *Given an infinite p-root cover $R \to S$ with R an integral perfectoid. Then the algebra*

$$Fil^\bullet_{\mathcal{N}} F^* \mathbb{\Delta}_{S/A} \in \mathcal{DF}_{(I,p)\text{-}comp}(Fil^\bullet_{\mathcal{N}} A)$$

is descendable.

Proof. Let us write $F^\bullet$ for the fibre of the map

$$Fil^\bullet_{\mathcal{N}} A \to Fil^\bullet_{\mathcal{N}} F^* \mathbb{\Delta}_{S/A}$$

and we claim that

$$\pi_0 \mathrm{Hom}_{\mathcal{DF}}((F^\bullet)^{\otimes 3}, Fil^\bullet_{\mathcal{N}} A) \simeq 0.$$

This will finish the proof by [BS17][11.20].

To check this, let us first make some reduction steps. First, we can take the tensor product over A. Also as the Nygaard filtration on A is complete, we can take the

completed tensor product and thus assume that both sides are complete. Now to see the above, by 3.6.7, it suffices to check that

$$\underline{\mathrm{Hom}}_{\mathcal{DF}}((F^\bullet)^{\otimes d+3}, Fil^\bullet_{\mathcal{N}} F^* A) \in \mathcal{DF}_{>1}$$

where we use the standard t-structure. As both sides are (assumed to be) complete, we can use 3.6.9 and the compatibility of taking associated graded with the internal Hom 3.6.8, to reduce this to showing that

$$\underline{\mathrm{Hom}}_{\mathcal{D}_{\mathrm{graded}}}(gr^\bullet(F^\bullet)^{\otimes d+3}, gr^\bullet_{\mathcal{N}} F^* A) \in (\mathcal{D}_{\mathrm{graded}})_{>2}.$$

Unwinding the internal Hom 3.6.8 and the definition of the t-structure, this boils down to showing that for each pair of integers i, j, we have

$$\mathrm{Hom}_{\mathcal{D}(R)}(gr^i(F^\bullet)^{\otimes 3}, gr^j_{\mathcal{N}} A) \in \mathcal{D}_{>2}.$$

For this, we recall from 3.1.5 that we have the following:

(a) The cofibre of the map $R \to S$ is a free R-module.
(b) $\mathcal{L}_{S/R}[-1]$ is a free S-module.

Now, let us identify the graded pieces with the filtered pieces in the conjugate filtration. Then using (b), we observe that the conjugate filtration on $\overline{\mathbb{\Delta}}_S$ takes the form

$$S \to \bigoplus_{I_1} S \to \bigoplus_{I_2} S \to \bigoplus_{I_3} S \to \dots$$

where each map is an inclusion of a direct summand. In particular using (b), we learn that

$$gr^i(F^\bullet) \simeq \bigoplus_K R[-1]$$

for some set K for all i. From this, one can easily observe what we want. □

We now come to the proof of 3.6.2(b).

Proof of 3.6.2(b). The assertion is stable under pullbacks and finite compositions. Thus, as explained in 3.1.4, we can assume that there is a cocartesian square

$$\begin{array}{ccc} \mathbf{Z}_p[x_i | i \in I] & \longrightarrow & \mathbf{Z}_p[x_i^{\frac{1}{p^\infty}} | i \in I] \\ \downarrow & & \downarrow \\ R & \longrightarrow & S. \end{array}$$

Furthermore, by lifting functions to a perfectoid mapping to R and pushing the upper horizontal map to R, we can assume R is perfectoid. By possibly refining the map, we also can assume B is semiperfectoid 3.1.8.

Using 3.3.13, we can identify the map in question with the map

$$\mathcal{R}(Fil^\bullet_{\mathcal{N}} \mathbb{\Delta}_S) \to \mathcal{R}(Fil^\bullet_{\mathcal{N}} \mathbb{\Delta}_R)$$

over $\mathbf{A}^1/\mathbf{G}_{\mathrm{m}}$. After pulling back along $\mathbf{A}^1 \to \mathbf{A}^1/\mathbf{G}_{\mathrm{m}}$, this map becomes a descendable map of algebras by 3.6.11.

To finish the proof, it now suffices to see that the map

$$\mathrm{Spa}(\oplus_i Fil^i_{\mathcal{N}} \mathbb{\Delta}_S) \to \mathrm{Spa}(\oplus_i Fil^i_{\mathcal{N}} \mathbb{\Delta}_R)$$

is proper. For this note first that for any $f \in Fil^i_{\mathcal{N}} \mathbb{\Delta}_S$ for $i \geq 1$, we have

$$\varphi(f) = f^p \bmod p \text{ and } \varphi(f) \in I.$$

This shows that such f are topologically nilpotent, and it suffices to see that the map

$$\mathbb{\Delta}_R \to \mathbb{\Delta}_S$$

is integral modulo (I, p). As this statement is stable under colimits in S, we can assume S arises as the pullback of a generic infinite p-root cover in one generator. In this case, using the conjugate filtration, it is easy to see that the map

$$S \to \bar{\mathbb{\Delta}}_S$$

is finite. □

3.7. **Solid Prismatic stacks.** We are now able to define the *Solid syntomification*.

Construction 3.7.1. Given a semiperfectoid S, as in the last section, we can associate to S the analytic stacks

$$S^{\mathcal{N},\square} \text{ and } S^{\mathbb{\Delta},\square} \simeq \mathrm{Spa}(\mathbb{\Delta}_S) \in \mathrm{AnStack}_{/\mathbf{Z}_p^{\mathbb{\Delta}}}.$$

Using 3.6.2 and 3.6.3, we see that this assignment sends infinite p-root covers and étale covers to effective epimorphisms of analytic stacks.

Now let us consider the topos

$$(\mathbf{Z}_p)_{\sqrt{\infty},\text{ét}}$$

generated by affine p-adic formal schemes via the union of the infinite p-root and the étale topology. Then this topos is generated by semiperfectoids 3.1.8. Using this observation, we obtain colimit-preserving functors

$$(_)^{\mathcal{N},\square} \text{ and } (_)^{\mathbb{\Delta},\square} \colon (\mathbf{Z}_p)_{\sqrt{\infty},\text{ét}} \to \mathrm{AnStack}_{/\mathbf{Z}_p^{\mathbb{\Delta}}}.$$

Note that $(_)^{\mathbb{\Delta},\square}$ also preserves finite limits, and that this also becomes true for $(_)^{\mathcal{N},\square}$ if we change the base in the target to

$$(\mathbf{Z}_p)^{\mathcal{N},\square}.$$

Definition 3.7.2. For a p-adic formal scheme X, we will call the analytic stack

$$X^{\mathbb{\Delta},\square}$$

the *solid Prismatisation* of X.

Analogously, we will refer to the analytic stack

$$X^{\mathcal{N},\square}$$

as the *solid Nygaard filtered Prismatisation* of X.

3.7.3. The open immersions from 3.3.12 induce disjoint open immersions

$$X^{\mathbb{\Delta},\square} \xrightarrow{j_{dR}} X^{\mathcal{N},\square} \xleftarrow{j_{HT}} X^{\mathbb{\Delta},\square}.$$

of analytic stacks. Again composing j_{dR} with the structure map recovers the identity, while composing j_{HT} with the structure map recovers the Frobenius.

Definition 3.7.4. Given a derived p-adic formal scheme X, we write $X^{\mathrm{syn},\square}$ for the pushout

$$\begin{array}{ccc} X^{\mathbb{\Delta},\square} \coprod X^{\mathbb{\Delta},\square} & \longrightarrow & X^{\mathcal{N},\square} \\ {\scriptstyle can}\downarrow & & \downarrow \\ X^{\mathbb{\Delta},\square} & \longrightarrow & X^{\mathrm{syn},\square} \end{array}$$

where the upper horizontal map is induced by the two open immersions j_{dR} and j_{HT}. We will refer to this analytic stack as the *solid Syntomification*.

3.7.5. Using that the open immersions j_{dR} and j_{HT} are stable under base change and descent for ∞-topoi, we see that for any morphism $f\colon X \to S$ of derived p-adic formal schemes, the induced square

$$\begin{array}{ccc} X^{\mathcal{N},\square} & \longrightarrow & S^{\mathcal{N},\square} \\ \downarrow & & \downarrow \\ X^{\mathrm{syn},\square} & \longrightarrow & S^{\mathrm{syn},\square} \end{array}$$

is Cartesian. Furthermore for any derived p-adic formal scheme X the map

$$X^{\mathcal{N},\square} \to X^{\mathrm{syn},\square}$$

is an étale surjection. Combining these two observations, one can deduce most properties for the solid syntomification from the solid Nygaard filtered Prismatisation.

3.7.6. For a p-adic formal scheme X, we can compute the category of solid prismatic F-gauges as the equalizer

$$\text{F-Gauge}^{\square}_{\mathbb{\Delta}}(X) \longrightarrow \mathcal{D}(X^{\mathcal{N},\square}) \underset{j_{HT}^*}{\overset{j_{dR}^*}{\rightrightarrows}} \mathcal{D}(X^{\mathbb{\Delta},\square}).$$

Let us record the following.

Proposition 3.7.7. *For a p-adic formal scheme X, the subcategories of dualizable objects canonically identify as*

- $\mathcal{D}(X^{\mathbb{\Delta},\square})^{dual} \simeq Perf(X^{\mathbb{\Delta}})$
- $\mathcal{D}(X^{\mathcal{N},\square})^{dual} \simeq Perf(X^{\mathcal{N}})$
- $F\text{-}Gauge^{\square}_{\mathbb{\Delta}}(X)^{dual} \simeq Perf(X^{Syn})$

where the categories on the right are the perfect complexes on the classical incarnations of these stacks as defined in [BL22b][6.1].

Proof. As taking the dualizable object of symmetric monoidal categories commutes with limits [Lur17][4.6.1.11], the claims easily reduce to the case $X \simeq \mathrm{Spf}(R)$ for R a semiperfectoid. In this case the claims follow from the next lemma. □

Lemma 3.7.8. Consider an animated ring A which is (derived) I-adically complete for some finitely generated ideal I together with some subring of integral elements $A^+ \subset A$. Then we have a canonical identification

$$\mathcal{D}(\mathrm{Spa}(A, A^+))^{\mathrm{dual}} \simeq \mathrm{Perf}(A).$$

PROOF. Let us first assume that A is discrete. Then any dualizable object is in particular nuclear and thus discrete by [And23][3.19]. But the dualizable objects in the classical derived category of a ring are exactly the perfect complexes. Now, to see the claim for general A, we first observe that for a dualizable object M the functor $M \otimes _$ preserves limits. Thus we have

$$M \simeq M \otimes A \simeq M \otimes \lim_n A/I^n \simeq \lim_n M/I^n$$

which shows that M is I-adically complete. From this we see that

$$\mathcal{D}(\mathrm{Spa}(A, A^+))^{\mathrm{dual}} \simeq \mathcal{D}(\mathrm{Spa}(A, A^+))^{\mathrm{dual}}_{\widehat{I}} \simeq \lim_n \mathcal{D}(\mathrm{Spa}(A/I^n, A^+))^{\mathrm{dual}} \simeq \mathrm{Perf}(A).$$

Where the second equivalence uses that taking duals commutes with limits [Lur17][4.6.1.11] and the third equivalence the case of a discrete ring. □

3.7.9. Recall from [BL22a][2.2.11+3.3.8] that there is a line bundle

$$\mathcal{O}_{\mathbf{Z}_p^{\mathbb{\Delta}}}\{1\} \in \mathrm{Perf}(\mathbf{Z}_p^{\mathbb{\Delta}})$$

called the *Breuil-Kisin twist*, which comes together with a frobenius isomorphism [BL22a][2.2.14]

$$\varphi\colon F^*\mathcal{O}_{\mathbf{Z}_p^{\mathbb{\Delta}}}\{1\} \simeq \mathcal{I}^{-1} \otimes \mathcal{O}_{\mathbf{Z}_p^{\mathbb{\Delta}}}\{1\}$$

where $\mathcal{I}$ denotes the Hode-Tate ideal. Heuristically, this line bundle should be thought of as

$$\otimes_{k\geq 0}(F^k)^*\mathcal{I}$$

where the Frobenius isomorphism is the evident one. Using 3.7.7, we can understand this object as a line bundle on $\mathbf{Z}_p^{\mathbb{\Delta},\square}$.

Recall that we had two maps

$$\mathbf{Z}_p^{\mathbb{\Delta},\square} \xleftarrow{\pi} \mathbf{Z}_p^{\mathcal{N},\square} \xrightarrow{t} \mathbf{D}^1_{\square}/(\mathbf{G}_{\mathrm{m}})^{\mathrm{an}}_{\square}$$

where π is the structure map and t is the Rees map. Using these maps, the *Nygaard filtered Breuil-Kisin twist* can be defined via

$$\mathcal{O}_{\mathbf{Z}_p^{\mathcal{N}}}\{1\} := \pi^*\mathcal{O}_{\mathbf{Z}_p^{\mathbb{\Delta}}}\{1\} \otimes t^*\mathcal{O}(-1).$$

We will understand this line bundle on $\mathbf{Z}_p^{\mathcal{N},\square}$.

Note that on the one hand we have

$$j_{dR}^*\mathcal{O}_{\mathbf{Z}_p^{\mathcal{N}}}\{1\} \simeq \mathcal{O}_{\mathbf{Z}_p^{\mathbb{\Delta}}}\{1\}$$

as composing j_{dR} with the structure map gives the identity, and the composition $t \circ j_{dR}$ factors over $(\mathbf{G}_{\mathrm{m}})^{\mathrm{an}}_{\square}/(\mathbf{G}_{\mathrm{m}})^{\mathrm{an}}_{\square}$. On the other hand, we have

$$j_{HT}^*\mathcal{O}_{\mathbf{Z}_p^{\mathcal{N}}}\{1\} \simeq F^*\mathcal{O}_{\mathbf{Z}_p^{\mathbb{\Delta}}}\{1\} \otimes \mathcal{I} \simeq \mathcal{O}_{\mathbf{Z}_p^{\mathbb{\Delta}}}\{1\}.$$

Where, for the first isomorphism, we use that composing j_{HT} with the structure map recovers the frobenius and that the composition $t \circ j_{HT}$ classifies the Hodge-Tate locus $\mathbf{Z}_p^{HT} \subset \mathbf{Z}_p^{\mathbb{\Delta}}$. The second Isomorphism comes from the Frobenius automorphism on the Breuil-Kisin twist recalled above. In particular, using this identification, we obtain a line bundle

$$\mathcal{O}_{\mathbf{Z}_p^{\mathrm{syn}}}\{1\}$$

on $\mathbf{Z}_p^{\mathrm{syn},\square}$ which we will refer to as the *syntomic Breuil-Kisin twist*.

By base change, we also obtain line bundles

$$\mathcal{O}_{X^{\mathbb{\Delta}}}\{1\}, \mathcal{O}_{X^{\mathcal{N}}}\{1\}, \mathcal{O}_{X^{\mathrm{syn}}}\{1\}$$

for an arbitrary derived p-adic formal scheme X.

3.7.10. Given a p-adic formal scheme X, then the mapping spectrum

$$\mathrm{Hom}_{X^{\mathrm{syn},\square}}(\mathcal{O}_{X^{\mathrm{syn},\square}}, \mathcal{O}_{X^{\mathrm{syn},\square}}\{n\})$$

via the presentation given in 3.7.6 become the fiber of the map

$$\varphi\{n\} - \mathrm{can}\colon Fil^n_{\mathcal{N}}\mathbb{\Delta}_X\{n\} \to \mathbb{\Delta}_X\{n\}$$

where $\varphi\{n\}$ is the twisted filtered frobenius and "can" comes from the inclusion of the filtration. In particular, this mapping spectrum identifies with the syntomic cohomology

$$R\Gamma_{\mathrm{syn}}(X, \mathbf{Z}_p(n))$$

for p-adic formal schemes as defined in [BL22a][7.4].

4. A six-functor formalism for syntomic cohomology

4.1. **Solid F-Gauges.** At this point, we have obtained a six-functor formalism for syntomic cohomology, of which we will now start to discuss properties.

Definition 4.1.1. Given a p-adic formal scheme X, we will write

$$\text{F-Gauge}^{\square}_{\mathbb{\Delta}}(X) := \mathcal{D}_{\mathrm{qc}}(X^{\mathrm{syn},\square})$$

for the category of quasi-coherent sheaves on the solid Syntomification of X and refer to it as *solid prismatic F-gauges* on X.

Construction 4.1.2. Recall from 2.1.10, that we have a six-functor formalism on analytic stacks, which considers the category of quasi-coherent sheaves on an analytic stack. Precomposing with the assignment

$$X \mapsto X^{\mathrm{syn},\square}$$

thus tells us that the assignment

$$X \mapsto \text{F-Gauge}^{\square}_{\mathbb{\Delta}}(X)$$

extends to a six-functor formalism.

The goal of the next two sections is to prove the following theorem.

Theorem 4.1.3. *The functor*

$$\textit{F-Gauge}^{\square}_{\mathbb{\Delta}}(_)\colon \textit{fSch}^{op}_{Spf(\mathbf{Z}_p)} \to \textit{Pr}^L_{st}$$

can be extended to a six-functor formalism on the category of (derived) p-adic formal schemes satisfying the following:

- *(A) Morphisms locally of finite type are !-able.*
- *(B) Étale morphisms are cohomologically étale.*
- *(C) Any finite composition of closed immersions and proper morphisms is weakly cohomologically proper.*
- *(D) The functor $(_)^{syn,\square}$ preserves étale covers and sends Zariski covers to open covers of analytic stacks. In particular the functors $\textit{F-Gauge}^{\square}_{\mathbb{\Delta}}(_)^*$ and $\textit{F-Gauge}^{\square}_{\mathbb{\Delta}}(_)^!$ are étale sheafs.*

(E) *It admits Tate twists. That is for any p-adic formal scheme X the object*

$$\mathcal{O}_{X^{syn}}(-1) := cof(\mathcal{O}_{X^{syn}} \to f_*\mathcal{O}_{(\mathbf{P}^1)^{syn}})$$

is $\otimes$-invertible and its inverse identifies with

$$\mathcal{O}_{X^{syn}}(1) \simeq \mathcal{O}_{X^{syn}}\{1\}[-2]$$

a shift of the Breuil-Kisin twist.

(F) *Any smooth morphism is cohomologically smooth. Furthermore, for such a smooth morphism $f\colon X \to S$, we have an identification*

$$f^!\mathcal{O}_{S^{syn}} := \omega_f \simeq \mathcal{O}_{X^{syn}}(d) \simeq \mathcal{O}_{X^{syn}}\{d\}[-2d]$$

of the dualizing sheaf, where d is the relative dimension of f.

(G) *The dualizable objects identify*

$$(F\text{-}Gauge^{\square}_{\mathbb{\Delta}}(_))^{dual} \simeq Perf(_)^{syn})$$

with perfect F-Gauges as defined in [BL22b][6.1]. In particular, there is a functorial identification

$$R\Gamma^{BMS}_{syn}(_, \mathbf{Z}_p(n)) \simeq Hom_{(_)^{syn}}(\mathcal{O}_{(_)^{syn}}, \mathcal{O}_{(_)^{syn}}\{n\})$$

of the mapping spectrum with the syntomic cohomology of p-adic formal schemes as defined in [BL22a].

Proof of 4.1.3(G). This was explained in 3.7.7 and 3.7.10. □

4.1.4. Note that assertions $(A), (B), (C)$ and (D) are étale local on the target. So using 3.7.5, we see that it suffices to check the analogous claims for the solid Nygaard filtered prismatisation. We will make use of this without further mentioning.

Proof of 4.1.3 (B) and (D). We claim that for an étale morphism (resp. an étale cover) $X \to S$ of p-adic formal schemes the induced morphism

$$X^{\mathcal{N},\square} \to S^{\mathcal{N},\square}$$

is an étale morphism (resp. an étale cover). As this assertion is local on the target, we can assume $S \simeq \mathrm{Spf}(R)$ for R a semiperfectoid. But then the claim is 3.6.2. □

Let us outsource parts of the proof in the following propositions.

Proposition 4.1.5. *Given a Zariski closed immersion $i\colon Z \to X$ of p-adic formal schemes. The induced morphism*

$$Z^{syn,\square} \to X^{syn,\square}$$

is affine and weakly proper.

Proof. It is enough to prove the analogous assertion for the solid Nygaard filtered prismatisation. Furthermore, using 4.1.3[(D)] we can assume that X (and thus also Z) is affine and then semiperfectoid. Let us write

$$\mathrm{Spf}(S) \simeq Z \to X \simeq \mathrm{Spf}(R).$$

As the assertion is stable under colimits in S, we can assume

$$S \simeq R/(f_1, \dots, f_n)$$

and by lifting the functions to a perfectoid mapping surjective to R, we can assume R to be perfectoid. Now one argues similarly to the last paragraph of the proof of 3.6.2[(c)]. □

Proposition 4.1.6. *Given a p-adic formal scheme X. The induced map*

$$(\mathbf{A}^n_X)^{syn,\square} \to X^{syn,\square}$$

is locally of $^+$finite type and thus in particular !-able.

Proof. As the assertion is stable under base change, we can assume $X \simeq \mathrm{Spf}(\mathbf{Z}_p)$. Now $\mathrm{Spf}(\mathbf{Z}_p)$ can be resolved by an integral perfectoid R via an infinite p-root cover. Thus we can assume $X \simeq \mathrm{Spf}(R)$ with R perfectoid.

Now write

$$\mathrm{Spf}(R^\infty) := \mathrm{Spf}(R[x_1^{\frac{1}{p^\infty}}, \dots, x_n^{\frac{1}{p^\infty}}]).$$

Then we have the infinite p-root cover

$$\mathrm{Spf}(R^\infty) \to \mathbf{A}^n_R$$

and it suffices to see that the map

$$(R^\infty)^{\mathrm{syn},\square} \to R^{\mathrm{syn},\square}$$

is of $^+$finite type. As in the proof of 3.6.2[(c)] for this it is enough to argue that the map

$$\mathrm{Spf}(\mathbb{A}_{\mathrm{inf}}(R^\infty)) \to \mathrm{Spf}(\mathbb{A}_{\mathrm{inf}}(R))$$

is of $^+$finite type. But as

$$\mathbb{A}_{\mathrm{inf}}(R^\infty) \simeq \mathbb{A}_{\mathrm{inf}}(R)[x_1^{\frac{1}{p^\infty}}, \dots, x_n^{\frac{1}{p^\infty}}]$$

this is clear. □

Proof of 4.1.3(A),(C). Using 4.1.3(D), (A) follows from 4.1.6 and 4.1.5.

We now prove (C). Consider a proper map $X \to S$, then the diagonal is a closed immersion, such that the induced map becomes weakly cohomologically proper by 4.1.5. Thus by 1.1.12 is suffices to see that the induced map is co-smooth.

By 4.1.3(D), we can assume S to be affine and resolving S by a semiperfectoid, we can even assume S to be semiperfectoid. As a proper map is of finite type, we can use 4.1.3(D), 4.1.6 and 4.1.5, to see that the induced map

$$X^{\mathcal{N},\square} \to S^{\mathcal{N},\square}$$

is locally of $^+$finite type. Thus by 2.3.3(E), we can check the assertion on a Zariski stratification of the target. Using the stratification 3.3.15 the claim now follows from 3.5.4 and 3.5.3. □

4.2. **The additive orientation.** To prove the rest, we will construct an additive orientation.

4.2.1. Using 3.7.10, we recall that there is a map

$$\mathrm{c}_1^{\mathrm{syn}} \colon R\Gamma_{\text{ét}}(X, \mathbf{G}_{\mathrm{m}})[-1] \to \mathrm{Hom}_{X^{\mathrm{syn}}}(\mathcal{O}_{X^{\mathrm{syn}}}, \mathcal{O}_{X^{\mathrm{syn}}}\{1\}[-2])$$

coming from the prismatic logarithm [BL22a][7.5.2]. This map is natural in X and identifies the target with the derived p-completion of the source [BL22a][7.5.6]. In particular, this gives us a theory of first Chern classes.

Construction 4.2.2. Using the first Chern classes from 4.2.1, as in 1.2.9 we can construct a morphism

$$\sum_{i=0}^{d} c_1^{\mathrm{syn}}(\mathcal{O}(1))^i\{d-i\}[2(i-d)]\colon \bigoplus_{i=0}^{d} \mathcal{O}_{\mathbf{Z}_p^{\mathrm{syn}}}\{d-i\}[2(i-d)] \to f_*\mathcal{O}_{(\mathbf{P}^d)^{\mathrm{syn}}}\{d\}[-2d]$$

where we write $f\colon \mathbf{P}^d \to \mathrm{Spf}(\mathbf{Z}_p)$ for the projection.

Proposition 4.2.3. *The morphism*

$$\sum_{i=0}^{d} c_1^{syn}(\mathcal{O}(1))^i\{d-i\}[2(i-d)]\colon \bigoplus_{i=0}^{d} \mathcal{O}_{\mathbf{Z}_p^{syn}}\{d-i\}[2(i-d)] \to f_*\mathcal{O}_{(\mathbf{P}^d)^{syn}}\{d\}[-2d]$$

is an isomorphism.

Proof. By resolving $\mathrm{Spf}(\mathbf{Z}_p)$ by a semiperfectoid (one can even find a perfectoid), we can assume that the base is given by $\mathrm{Spf}(R)$ for R a semiperfectoid. Using the formula of mapping spectra recalled in 3.7.7, it suffices to check that the maps

(a) $\bigoplus_{i=0}^{d} \mathcal{O}_{R^{\mathbb{\Delta}}}\{d-i\}[2(i-d)] \to f_*\mathcal{O}_{(\mathbf{P}^d)^{\mathbb{\Delta}}}\{d\}[-2d]$ in $\mathcal{D}(R^{\mathbb{\Delta}})$
(b) $\bigoplus_{i=0}^{d} \mathcal{O}_{R^{\mathcal{N}}}\{d-i\}[2(i-d)] \to f_*\mathcal{O}_{(\mathbf{P}^d)^{\mathcal{N}}}\{d\}[-2d]$ in $\mathcal{D}(R^{\mathcal{N}})$

are isomorphisms. As the lower-$*$ functor commutes with limits, both sides of these maps are (p,I)-adically complete. Thus we can check these isomorphisms modulo (I,p). We will do so but omit the modulo notation in the following. Now in $\mathcal{D}(R^{\mathbb{\Delta},\square})$ mapping out of a free module on a profinite set $S \simeq \lim_n S_n$ is conservative and similar mapping out of shifted versions of such a free guy is conservative on $\mathcal{D}(R^{\mathcal{N},\square})$. We now claim that on both sides of the map we can compute mapping out of such a free guy by

$$\mathrm{Hom}_{\mathcal{D}(R^{\mathbb{\Delta},\square})}(S,_) \simeq \mathrm{colim}_n \mathrm{Hom}_{\mathcal{D}(R^{\mathbb{\Delta},\square})}(S_n,_)$$

and similar for $\mathcal{D}(R^{\mathcal{N},\square})$. For the domains of the maps, that is clear. For the targets, we can first choose the standard Zariski presentation of projective spaces, which writes the target as a finite limit of cohomologies of affine spaces and circles. Those we can now resolve by semiperfectoids, and the claim follows as filtered colimits commute with totalisations in non-connective spectra. Using this formula, the claims now follow from [BL22a][9.1.4.(4)] and [BL22a][9.1.4.(5)] respectively. □

4.2.4. On the category of (derived) p-adic formal schemes, the collection of étale and smooth morphisms define a class of geometrically étale and smooth morphisms in the sense of 1.3.3. To construct blow-ups, we can locally p-adically complete the construction from (derived) algebraic geometry [KR25] [Tan26][A] and then glue. All the other assertions in the definition are clear.

Proposition 4.2.5. *F-Gauge$^{\square}_{\mathbb{\Delta}}$-cohomology satisfies blow-up excision in the sense of 1.5.2.*

PROOF. We have to see that the square

$$\begin{array}{ccc} \mathbf{P}^{n-1}_{\mathbf{Z}_p} & \longrightarrow & \mathbf{V}_{\mathbf{P}^{n-1}_{\mathbf{Z}_p}}(\mathcal{O}(1)) \\ \downarrow & \overset{f}{\searrow} & \downarrow p \\ \{0\} & \xrightarrow[0]{} & \mathbf{A}^n_{\mathbf{Z}_p} \end{array}$$

induces a (co)cartesian square on cohomology in $\text{F-Gauge}^{\square}_{\mathbb{\Delta}}(\mathbf{A}^n_{\mathbf{Z}_p})$. Using proper base change, we can check this after pulling back along an infinite p-root cover. Thus by resolving $\mathrm{Spf}(\mathbf{Z}_p)$ by a semiperfectoid $\mathrm{Spf}(R)$ and then resolving $\mathbf{A}^n_R$ by a semiperfectoid, it suffices to check the analogous claim for a blow-up square

$$\begin{array}{ccc} E & \longrightarrow & B \\ \downarrow & & \downarrow \\ Z & \longrightarrow & \mathrm{Spf}(R) \end{array}$$

where R is semiperfectoid (even perfectoid if the reader likes). Note that such resolutions exist in such a way that all p-adic formal schemes in the square are the p-adic completions of classical schemes. Now, similar to the proof of 4.2.3 the claim follows from [BL22a][9.4.6]. □

Proof of 4.1.3(E),(F). Assertion (E) easily follows from the projective bundle formula 4.2.3. To deduce assertion (F), we note that the construction of first Chern classes 4.2.1 together with the projective bundle formula 4.2.3 and the class of étale and smooth morphisms 4.2.4 give an additively oriented six-functor formalism (for the other assertions we use the rest of 4.1.3). Furthermore, it satisfies blow-up excision by 4.2.5. Thus we can apply 1.6.5 and 1.6.6. □

This finishes the proof of 4.1.3.

4.3. **Realizations.** We will now discuss some realizations.

4.3.1. We consider the following situation. Assume we have given an analytic stack $\mathcal{B}$ together with a map

$$\mathcal{B} \to \mathbf{Z}_p^{\mathrm{syn}\square}.$$

Then we automatically obtain a well-behaved six-functor formalism for p-adic formal schemes by the following construction.

For a p-adic formal scheme X, we first consider the cartesian square

$$\begin{array}{ccc} X_{\mathcal{B}} & \longrightarrow & X^{\mathrm{syn},\square} \\ \downarrow & & \downarrow \\ \mathcal{B} & \longrightarrow & \mathbf{Z}_p^{\mathrm{syn},\square} \end{array}$$

and then associate to X the category

$$\mathcal{D}(X_{\mathcal{B}}).$$

As all the assertions (A)-(F) of 4.1.3 are stable under base change, this six-functor formalism will satisfy analogous assertions, where one defines the Tate-twist by base change. We will now discuss some examples of this phenomenon.

Example 4.3.2. For any p-adic formal scheme, we have a cartesian square

$$\begin{array}{ccc} X^{\mathbb{\Delta},\square} & \xrightarrow{j_{dR}} & X^{\mathrm{syn},\square} \\ \downarrow & & \downarrow \\ \mathbf{Z}_p^{\mathbb{\Delta},\square} & \xrightarrow[j_{dR}]{} & \mathbf{Z}_p^{\mathrm{syn},\square}. \end{array}$$

In particular, using the deRham point in 4.3.1, we obtain a well-behaved six-functor formalism for Prismatic cohomology.

Example 4.3.3. Similar to the last example, we can use the map

$$\mathbf{Z}_p^{\mathrm{HT},\square} \to \mathbf{Z}_p^{\mathrm{syn},\square}$$

to obtain a six-functor formalism for Hodge-Tate cohomology.

Construction 4.3.4. Recall that we had the structure map

$$\mathbf{Z}_p^{\mathbb{\Delta},\square} \to \mathbf{D}^1/\,{\mathbf{G}_{\mathrm{m}}}^{\mathrm{an}}$$

which came from base changing a Cartier-Witt divisor along the map $W \to \mathbf{G}_{\mathrm{a}}$. Note that the action of ${\mathbf{G}_{\mathrm{m}}}^{\mathrm{an}}$ restricts to the substack

$$\mathrm{colim}_n \,\mathrm{Spa}(\mathbf{Z}_p[x]/(x^n)) \subset \mathbf{D}^1$$

such that we can base change this structure map along this map quotient by ${\mathbf{G}_{\mathrm{m}}}^{\mathrm{an}}$. Further base changing along the inclusion

$$\mathrm{colim}_n \,\mathrm{Spa}(\mathbf{Z}/p^n) \to \mathrm{Spa}(\mathbf{Z}_p)$$

we obtain an (open) substack

$$\mathbf{Z}_p^{\mathbb{\Delta}} \subset \mathbf{Z}_p^{\mathbb{\Delta},\square}.$$

To also modify the solid Nygaard filtered prismatisation, we consider the cartesian square

$$\begin{array}{ccc} \mathbf{Z}_p^{\mathcal{N}} & \longrightarrow & \mathbf{Z}_p^{\mathcal{N},\square} \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ \mathbf{Z}_p^{\mathbb{\Delta}} & \longrightarrow & \mathbf{Z}_p^{\mathbb{\Delta},\square}. \end{array}$$

And finally, analogously to the solid Syntomification, we construct the pushout

$$\begin{array}{ccc} \mathbf{Z}_p^{\mathbb{\Delta}} \amalg \mathbf{Z}_p^{\mathbb{\Delta}} & \longrightarrow & \mathbf{Z}_p^{\mathcal{N}} \\ \downarrow & & \downarrow \\ \mathbf{Z}_p^{\mathbb{\Delta}} & \longrightarrow & \mathbf{Z}_p^{\mathrm{syn}}. \end{array}$$

Note that we have a map $\mathbf{Z}_p^{\mathrm{syn}} \to \mathbf{Z}_p^{\mathrm{syn},\square}$.

Definition 4.3.5. For a p-adic formal scheme X, we define the *classical Syntomification* to be the fiber product

$$\begin{array}{ccc} X^{\mathrm{syn}} & \longrightarrow & X^{\mathrm{syn},\square} \\ \downarrow & & \downarrow \\ \mathbf{Z}_p^{\mathrm{syn}} & \longrightarrow & \mathbf{Z}_p^{\mathrm{syn},\square}. \end{array}$$

We now obtain the following corollary.

Theorem 4.3.6. *The functor*

$$\mathcal{D}((_)^{syn})\colon fSch^{op}_{Spf(\mathbf{Z}_p)} \to Pr^L$$

can be extended to a six-functor formalism on the category of (derived) p-adic formal schemes satisfying (A)-(F) of 4.1.3. Furthermore, assertion (G) can be extended to: There is a symmetric monoidal and fully faithful natural transformation

$$F\text{-}Gauge_{\mathbb{\Delta}}(_) \to \mathcal{D}((_)^{syn})$$

where the left-hand side denotes the category of prismatic F-Gauges as defined in [BL22b].

Proof. As explained in 4.3.1, the first assertion follows from 4.1.3. To prove the second one, by descent, it suffices to evaluate on a semiperfectoid R. Then, using the analogous equalizer diagrams as in 3.7.6, it suffices to check the analogous claims for

$$R^{\mathcal{N}} \text{ and } R^{\mathbb{\Delta}}.$$

Note that $\mathcal{D}(R^{\mathcal{N}})$ identifies with the subcategory of $\mathcal{D}(R^{\mathcal{N},\square})$ consisting of $(p.I)$-adically complete objects. Thus the claim reduces to the assertion that for an animated ring A the functor

$$\mathcal{D}(A)^{\mathrm{cl}} \to \mathcal{D}(\mathrm{Spa}(A)),$$

where we denote by $\mathcal{D}(_)^{\mathrm{cl}}$ the classical derived category, is symmetric monoidal and fully faithful. □

Let us now consider an étale realization. For this we will work over $\mathrm{Spf}(\mathbf{Z}_p^{\mathrm{cyc}})$.

Construction 4.3.7. Recall from [BL22a][8.5] that choosing a compatible system of p-th power roots of unity $(1, \xi_p, \xi_p^2, \dots)$ in the Tate module of $\mathbf{Z}_p^{\mathrm{cyc}}$ determines a map

$$\epsilon\colon \mathcal{O}_{(\mathbf{Z}_p^{\mathrm{cyc}})^{\mathrm{syn}}}\{-1\} \to \mathcal{O}_{(\mathbf{Z}_p^{\mathrm{cyc}})^{\mathrm{syn}}}.$$

Thus from this we obtain a section

$$v_1 = \epsilon^{p-1}\colon (\mathbf{Z}_p^{\mathrm{cyc}})^{\mathrm{syn},\square} \to \overline{\mathbf{V}(\mathcal{O}\{-(p-1)\})}^{10}.$$

Looking at the algebraic non-vanishing locus[11] of this section and further base changing along $\mathrm{colim}_n \mathrm{Spa}(\mathbf{Z}/p^n) \to \mathrm{Spa}(\mathbf{Z}_p)$, we obtain a map

$$(\mathbf{Q}_p^{\mathrm{cyc}})^{\text{ét},\square} \to (\mathbf{Z}_p^{\mathrm{cyc}})^{\mathrm{syn},\square}.$$

Definition 4.3.8. For a p-adic formal scheme X over $\mathbf{Z}_p^{\mathrm{cyc}}$, we will refer to the fibre product

$$\begin{array}{ccc} X_\eta^{\text{ét},\square} & \longrightarrow & X^{\mathrm{syn},\square} \\ \downarrow & & \downarrow \\ (\mathbf{Q}_p^{\mathrm{cyc}})^{\text{ét},\square} & \longrightarrow & (\mathbf{Z}_p^{\mathrm{cyc}})^{\mathrm{syn},\square} \end{array}$$

as the *étale locus* of the solid Syntomification of X.

[10] The vector bundle here is interpreted purly algebraically. In particular, it is proper.

[11] Also this is a closed substack. That is, locally we just invert an element algebraically.

4.3.9. Let us compute how $R_\eta^{\text{ét},\square}$ looks like for R a semiperfectoid. First, using [BL22a][8.5.3+2.6.1], we see that the fibre product

$$\begin{array}{ccc} Q & \longrightarrow & R^{\mathcal{N},\square} \\ \downarrow & & \downarrow \\ (\mathbf{Q}_p^{\text{cyc}})^{\text{ét},\square} & \longrightarrow & (\mathbf{Z}_p^{\text{cyc}})^{\text{syn},\square} \end{array}$$

is given by

$$Q \simeq \operatorname{colim}_n \operatorname{Spa}(\mathbb{\Delta}_R[\frac{1}{I}]/p^n, \mathbb{\Delta}_R)$$

and we also get the same formula for the Prismatisation. As we identify the Frobenius with the identity, we thus obtain the following pushout formula

$$\begin{array}{ccc} Q^{\text{perf}} \amalg Q^{\text{perf}} & \xrightarrow{(id,\varphi)} & Q^{\text{perf}} \\ {\scriptstyle can}\downarrow & & \downarrow \\ Q^{\text{perf}} & \longrightarrow & R_\eta^{\text{ét},\square} \end{array}$$

where we write

$$Q^{\text{perf}} \simeq \operatorname{colim}_n \operatorname{Spa}(\mathbb{\Delta}_{R,\text{perf}}[\frac{1}{I}]/p^n, \mathbb{\Delta}_{R,\text{perf}})$$

using perfect prismatic cohomology [BS19][7+8].

We now obtain the following.

Theorem 4.3.10. *The functor*

$$\mathcal{D}((_)_\eta^{ét,\square})\colon fSch^{op}_{/Spf(\mathbf{Z}_p^{cyc})} \to Pr^L$$

extends to a six-functor formalism on p-adic formal schemes over $\mathbf{Z}_p^{cyc}$*, which satisfies analogous assertions to (A)-(F). Furthermore, the dualizable objects naturally identify*

$$\mathcal{D}((_)_\eta^{ét,\square})^{dual} \simeq \mathcal{D}^b_{lisse}((_)_\eta, \mathbf{Z}_p)$$

with the category of lisse étale sheaves on the generic fiber.

Proof. As explained in 4.3.1, the first assertion follows from 4.1.3. To compute the dualizable objects, we first claim that it is enough to construct this equivalence on integral perfectoids. On the left, this follows from descending to semiperfectoids and then using 4.3.9. For the right, this follows from arc-descent [BM21][5.13] for constructible étale sheaves together with the fact that perfectoids form a basis for the arc-topology [BS19][8.8].

Now using [And21][5.50] and 4.3.9, we see that on a perfectoid R the left-hand side is computed by

$$\operatorname{Perf}(\mathbb{A}_{\text{inf}}(R)[\frac{1}{d}]_{\widehat{p}})^{\varphi=id}.$$

Thus the claim follows from Tilting and the Artin-Schreier sequence (see [BS21][3.7]). □

Remark 4.3.11. If one is happy to consider $\mathbf{F}_p$-coefficients, 4.3.10 can be extended to all p-adic formal schemes [BM22][2.7].

Remark 4.3.12. The étale realization considered here is somewhat suboptimal. For example

$$\mathcal{D}((_)_\eta^{\text{ét},\square})$$

does not satisfy excision[12] and thus cannot capture all étale sheaves. This issue will be addressed elsewhere.

References

[AHI24] Toni Annala, Marc Hoyois, and Ryomei Iwasa. Atiyah duality for motivic spectra. *preprint arXiv:2403.01561*, 2024.

[AHI25] Toni Annala, Marc Hoyois, and Ryomei Iwasa. Algebraic cobordism and a Conner-Floyd isomorphism for algebraic K-theory. *Journal of the American Mathematical Society*, 38(1):243–289, arXiv:2303.02051 2025.

[And21] Grigory Andreychev. Pseudocoherent and Perfect Complexes and Vector Bundles on Analytic Adic Spaces. *preprint arXiv:2105.12591*, 2021.

[And23] Grigory Andreychev. K-theorie adischer Räume. *preprint arXiv:2311.04394v1*, 2023.

[BK26] Tess Bouis and Arnab Kundu. $\mathbb{A}^1$-connectivity of motivic spaces. *Journal of the Institute of Mathematics of Jussieu*, pages 1–11, arXiv:2512.10712v2 2026.

[BL22a] Bhargav Bhatt and Jacob Lurie. Absolute prismatic cohomology. *preprint arXiv:2201.06120*, 2022.

[BL22b] Bhargav Bhatt and Jacob Lurie. Prismatic F-gauges (Lecture notes). math.ias.edu/ bhatt/teaching/mat549f22/lectures.pdf, 2022.

[BL22c] Bhargav Bhatt and Jacob Lurie. The prismatization of p-adic formal schemes. *preprint arXiv:2201.06124*, 2022.

[BM21] Bhargav Bhatt and Akhil Mathew. The arc-topology. *Duke Mathematical Journal*, 170(9):1899–1988, arXiv:1807.04725v4 2021.

[BM22] Bhargav Bhatt and Akhil Mathew. Syntomic complexes and p-adic étale Tate twists. *Forum of Mathematics, Pi*, 11, arXiv:2202.04818v2 2022.

[BMS18] Bhargav Bhatt, Matthew Morrow, and Peter Scholze. Topological Hochschild homology and integral p-adic Hodge theory. *Publications mathématiques de l'IHÉS*, 129:199–310, arXiv:1802.03261 2018.

[Bos23] Guido Bosco. Rational p-adic Hodge theory for rigid-analytic varieties. *preprint arXiv:2306.06100*, 2023.

[BS17] Bhargav Bhatt and Peter Scholze. Projectivity of the Witt vector affine Grassmannian. *Inventiones mathematicae*, 209(2):329–423, arXiv:1507.06490 2017.

[BS19] Bhargav Bhatt and Peter Scholze. Prisms and prismatic cohomology. *Annals of Mathematics*, arXiv:1905.08229 2019.

[BS21] Bhargav Bhatt and Peter Scholze. Prismatic F-crystals and crystalline Galois representations. *Cambridge Journal of Mathematics*, arXiv:2106.14735 2021.

[Cam24] Juan Esteban Rodríguez Camargo. The analytic de Rham stack in rigid geometry. *preprint arXiv:2401.07738*, 2024.

[CD09] Denis-Charles Cisinski and Frédéric Déglise. Triangulated Categories of Mixed Motives. *Springer Monographs in Mathematics*, arXiv:0912.2110 2009.

[CS19] Dustin Clausen and Peter Scholze. Lectures on Condensed Mathematics. *preprint arXiv:2605.03658v1*, May 2019.

[CS22] Dustin Clausen and Peter Scholze. Condensed Mathematics and Complex Geometry. *preprint arXiv:2605.11731v1*, 2022.

[CS24] Dustin Clausen and Peter Scholze. Analytic Stacks Lecture (Lecture). youtube, 2024.

[CTHK97] Jean-Louis Colliot-Thélène, Raymond T. Hoobler, and Bruno Kahn. The Bloch-Ogus-Gabber theorem. In *Algebraic K-Theory (Toronto, ON, 1996)*, volume 16, pages 31–94. American Mathematical Society, 1997.

[DK14] Christopher Davis and Kiran S. Kedlaya. On the Witt vector Frobenius. *Proceedings of the American Mathematical Society*, 142(7):2211–2226, arXiv:1409.7530 2014.

[Dri24] Vladimir Drinfeld. Prismatization. *Selecta Mathematica*, 30(3):49, arXiv:2005.04746 2024.

[12] This is also known as Localization or Kashiwara's lemma.

[GM24] Zachary Gardner and Keerthi Madapusi. An algebraicity conjecture of Drinfeld and the moduli of p-divisible groups. *preprint arXiv:2412.10226v4*, 2024.

[GP16] Owen Gwilliam and Dmitri Pavlov. Enhancing the filtered derived category. *Journal of Pure and Applied Algebra*, 222(11):3621–3674, arXiv:1602.01515 2016.

[GR17] Dennis Gaitsgory and Nick Rozenblyum. *A Study in Derived Algebraic Geometry, Volume I: Correspondences and Duality*, volume 1 of *Mathematical Surveys and Monographs*. American Mathematical Society, 2017.

[Hau25] Maximilian Hauck. An arithmetic étale-crystalline comparison with coefficients in crystalline local systems. *preprint arXiv:2505.03294v1*, 2025.

[HM24] Claudius Heyer and Lucas Mann. 6-Functor Formalisms and Smooth Representations. *preprint arXiv:2410.13038*, 2024.

[Hub93] R. Huber. Continuous valuations. *Mathematische Zeitschrift*, 212(1):455–477, 1993.

[Hub96] Roland Huber. *Étale Cohomology of Rigid Analytic Varieties and Adic Spaces*, volume 30 of *Aspects of Mathematics*. Vieweg+Teubner Verlag, 1996.

[KH94] Manfred Knebusch and Roland Huber. On valuation spectra. *Contemporary Mathematics*, 155, 01 1994.

[KR25] Adeel A. Khan and David Rydh. Virtual Cartier divisors and blow-ups. *Selecta Mathematica*, 31(4):1–28, arXiv:1802.05702 2025.

[LM25] Dhilan Lahoti and Deven Manam. Cohomology theories in the moduli of ring stacks. *preprint arXiv:2510.09582v2*, 2025.

[Lur] Jacob Lurie. Kerodon. .

[Lur17] Jacob Lurie. Higher Algebra. math.ias.edu/ lurie/papers/HA.pdf 2017.

[Man22] Lucas Mann. *A p-Adic 6-Functor Formalism in Rigid-Analytic Geometry*. PhD thesis, Rheinische Friedrich-Wilhelms-Universitaet Bonn (Germany), arXiv:2206.02022 2022.

[Mat16] Akhil Mathew. The Galois group of a stable homotopy theory. *Advances in Mathematics*, 291:403–541, arXiv:1404.2156 2016.

[MM25] Akhil Mathew and Shubhodip Mondal. Affine stacks and derived rings. https://www.math.purdue.edu/~mondalsh/papers/affinestacks.pdf, 06 2025.

[Mou19] Tasos Moulinos. The geometry of filtrations. *Bulletin of the London Mathematical Society*, 53(5):1486–1499, arXiv:1907.13562 2019.

[MV99] Fabien Morel and Vladimir Voevodsky. $\mathbb{A}^1$-homotopy theory of schemes. *Publications Mathématiques de l'Institut des Hautes Études Scientifiques*, 90(1):45–143, 1999.

[Sch25] Peter Scholze. Six-Functor Formalisms. *preprint arXiv:2510.26269v2*, 2025.

[Sta] The Stacks Project Authors Stacks. Stacks Project. https://stacks.math.columbia.edu.

[Tan] Longke Tang. Gysin maps for oriented $\mathbb{P}^1$-spectra. Link.

[Tan26] Long Tang. The $\mathbb{P}^1$-motivic gysin map. *preprint arXiv:2604.24888v2*, 2026.

[Wed19] Torsten Wedhorn. Adic spaces. *preprint arXiv:1910.05934*, 2019.

[Zav23] Bogdan Zavyalov. Poincaré Duality in abstract 6-functor formalisms. *preprint arXiv:2301.03821*, 2023.

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